



POLITECNICO MILANO 1863

Requirements Analysis Specification Document

Ali Noshad, ali.noshad@mail.polimi.it, 101108

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Students&Companies - Requirements Analysis Specification
Document

Prof. Matteo Rossi

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1 Introduction

The Requirement Analysis and Specification Document (RASD) aims to focus on the tasks needed to develop and implement an application, taking account of the requirements of the involved stakeholders and, analyzing and documenting also the application requirements. Then, in the second part of the document, there is a more formal definition of the requirements with the use of the Alloy language. In general this document is meant for developers tasked with the implementation of the system and also for all the other entities involved in validation and managing of the project.

1.1 Purpose

Recently job market becoming more and more competitive. Accordingly students face significant challenges in finding suitable job opportunities and internships with respect to their experiences, skills and most importantly their interest and their career goals. Simultaneously, companies find it hard who with fit their need and requirements as well as their work culture. The Students&Companies (S&C) platform addresses this need by streamlining and enhancing the process of matching students with relevant internship opportunities. By analyzing the experiences, skills and attitudes of students listed in their resume and comparing them with the specific requirements and perks of available roles provided by the companies, S&C connects students with positions that suit them well.

The platform helps students easily find internship opportunities by collecting detailed information about available internships, student resumes, skills, and preferences. This allows students to search for companies and roles that match their interests through S&C's search system, giving them more control over finding an internship. Additionally, S&C assists companies by managing interview schedules and assessments for students who apply for specific roles, making the hiring process smoother and helping both sides make better decisions. Finally, S&C offers tools to track internships by gathering student feedback and complaints, providing insights to improve future programs. Overall, S&C aims to connect students and companies, making it easier and more transparent for students to start their careers.

This document focuses on the Requirements Analysis and Specification Document (RASD) of the system and describes the main goals, the domain assumptions, the scenarios which may happen, the uses cases, the list of functional and non-functional requirements which system should fulfill, and finally the diagrams to visualize the interactions between components and performance of the system.

1.1.1 Goals

Table 1: Goals Definition

Goals	Description
G1	S&C allows students to manage (create, upload, edit) their resume
G2	S&C allows students to see available internships (e.g., by location, industry)
G3	S&C allows students to initiate the search process and submit applications directly
G4	S&C allows students to track the status (reviewed, shortlisted, selected) of their internship applications
G5	S&C allows students to raise complaints or concerns regarding an internship experience
G6	S&C allows companies to manage (create, edit, publish, delete) internship listings
G7	S&C allows companies to browse and search through student profiles (resume) applied for the internship (e.g., by location, industry)
G8	S&C allows companies the ability to shortlist candidates
G9	S&C allows companies communicate with applicants
G10	S&C allows companies to provide updates on their application status of incoming applications
G11	S&C allows companies to manage (create, edit, send) questionnaire to applicants
G12	S&C allows companies to manage (scheduling and conducting) interviews with applicants
G13	S&C allows universities to track application processes
G14	S&C allows universities to monitor internship execution
G15	S&C allows universities to collect complaints

1.2 Scope

The Students & Companies (S&C) platform connects and facilitate matching university students want to find internships with companies providing those opportunities. The platform will provide abilities for students to manage their profiles, apply for internships, and track their application status. At the same time, it will allow companies to post internships offerings, review applications, and communicate with potential candidates. The platform also includes an access for universities to monitor internship activities and collect compliance.

To initiate application process, an application has been defined through which the user can do the following:

Student Features

- Students must register to system with their credentials.
- Student must log in to his/her account. Then, students can create, upload, and edit their resumes.

- Then, students can view available internships and filter them based on various criteria.
- After selecting desired internships, students can initiate applications for internships and track the status of their applications (e.g., reviewed, shortlisted, selected).
- Furthermore, students can raise complaints or concerns regarding their internship experiences.

Company Features

- Companies must register to system with their credentials.
- Companies must log in to their account, and then, an create, edit, publish, and delete internship listings with all the necessary information (e.g. requirements, details of the job, ...)
- Then, for each internship posting, companies can browse and search through student profiles (resumes) that have applied for their internships.
- After receiving applications, companies can review incoming applications and shortlist candidates based on their qualifications.
- Then, companies can communicate with applicants and provide updates on application statuses.
- Companies can schedule and conduct interviews with shortlisted candidates and manage questionnaires for applicants.
- Finally, companies can decide which applicants are suitable for the positions based on interview and questionnaire.

Universities Features

- universities must register and login to their account.
- Then, universities can monitor the application processes of their students.
- Also, universities can oversee the execution of internships to ensure compliance and quality.

1.2.1 World Phenomena

Table 2: World Phenomena

World Phenomena	Description
WP1	Students actively look for internship opportunities based on their interest and skills and knowledge
WP2	Students prepare resume
WP3	Companies offer various internships with specific requirements, expectations, and work cultures, which are of interest to potential student applicants.
WP4	Companies review students resume
WP5	Companies decision on short listing candidates
WP6	Companies decision on acceptance/rejection of candidates
WP7	Students accept/reject the offer of internship
WP8	Students start the internship

1.2.2 Shared Phenomena

Table 3: Shared Phenomena

Shared phenomena	Description	Control
SP1	Students create, upload, and edit their resumes within the system to apply for internships	World
SP2	Companies create, publish, edit, and delete internship listings through the platform	World
SP3	System shows the available internships	System
SP4	Students use the system's search and filtering functions to explore available internship opportunities	World
SP5	Students apply to the internship through the system	World
SP6	Companies search and filter applicants through the system	World
SP7	Companies manage (create, edit, publish, delete) questionnaire through the system	World
SP8	Companies schedule a meeting through the system	World
SP9	Students are notified about a change in applications	System
SP10	System shows all the students applied for an internship	System
SP11	System notifies the universities about the status and process of internship	System
SP12	System notifies the companies about whether the internship accepted/rejected	System

1.3 Definitions, Acronyms, Abbreviations

Table 4: Abbreviations

Abbreviations	Description
RASD	Requirements Analysis & Specification Document
G	Goal
R	Requirement
C	Component
WP	World Phenomena
SP	Shared Phenomena
S&C	Students and Companies

1.4 Revision history

- Version 1.0 - 1/12/2024

1.5 Reference Documents

- Specification Document: “Assignment RDD AY 2024-2025.pdf”
- Course Slides
- IEEE Standard Documentation for RASD

1.6 Document Structure

Section 1 Introduces the problem and the purpose of the project and defining the scope of the Students&Companies. Describe the specifications such as the definitions, acronyms, abbreviations, revision history, and references. As well as introducing the goals, world and shared phenomena of the software.

Section 2 Identifying key scenarios and detailing the software’s primary features through class diagrams and state charts. The user characteristics section describes the types of actors interacting with the application, while the product function subsection outlines its functionalities. Finally, domain assumptions are established.

Section 3 The main part of the project which introduces interface requirements such as user interface, hardware interface, software interface, and communication interfaces. Presenting the functional requirements that are shown by use case diagrams and sequence diagrams. Then requirements are mapped to use cases.

Section 4 Using Alloy language for analyzing the system and brief comments for clarifying the Alloy codes.

Section 5 Shows how much time is spent on this project.

Section 6 Contains the references

2 Overall Description

2.1 Product perspective

2.1.1 Scenarios

Student Uploads Resume Mia is sitting at her desk at the university library, she is excited to showcase her skills, capabilities, and experience on the S&C platform. She navigates to her profile page, where she finds an option to upload her Resume. Mia selects “Add Resume”, which opens a file explorer window. With precision, she locates her Resume file within the organized “Career” folder on her desktop. She clicks to upload the document. The platform start the upload process, automatically checking the document format to ensure compatibility. A progress indicator shows the confirmation of the upload, and once completed, Mia can review her uploaded Resume to confirm accuracy and ensure it aligns with her career goals and information.

Company Posts Internship Alex is a representative of company called BigTech and working in hiring manage department. She has been notified by her boss that a new intern is required for R&D department. Accordingly, Alex is preparing to post a new internship. He logs into the platform, selects “Post Internship,” and begins to fill out the details. He describes the responsibilities, the technical skills required, and any perks like mentorship or training programs. After adding all the information, Alex hits “Submit.” Within moments, Mia and other students can now view the new opportunity and consider applying.

Viewing internship opportunities On the S&C platform, Emma uses the search feature to find internships by entering keywords like “machine learning,” “AI,” and “data analysis.” She filters results by location, seeking options close to her university campus, and sorts the list by the latest postings. After browsing several listings, Emma finds the TechVision Corp internship. she clicks in to learn

more, reading through the description to see if it aligns with her career aspirations. She reviews the details of the position, compares it with other internships, and decides it's a good match.

Student Initiates Application After reading through the details, Mia feels Alex's internship is a perfect match for her goals. She clicks "Apply," and her Resume is immediately sent to Alex's company profile. The platform confirms her application, and Mia's dashboard updates to show the internship as "Applied." She's thrilled to take her first step towards the role.

Company views all the applicants for internship posting Sarah, an HR manager at TechVision Corp, has posted an internship for a machine learning role on the S&C platform. After several days, she has received numerous applications. Sarah logs into the S&C platform and navigates to the "Internships" dashboard for her internship posting. The platform provides a list of applicants, along with their Resume. Sarah uses the platform's filtering options to shortlist applicants based on criteria such as relevant skills and project experience, enabling her to focus only on the most qualified candidates.

Scheduling Interviews by company's hiring manager After reviewing the applications, Sarah decides that Emma and a few other candidates are well-suited for the role. She wants to set up interviews to get to know them better. Sarah selects Emma's profile and clicks "Schedule Interview" on the S&C platform. A calendar view appears, showing Emma's availability based on her provided schedule, along with suggested time slots to minimize conflicts. Sarah picks an available slot and sends the interview invitation. Emma receives a notification on the S&C platform, confirming the interview time and including a link to join the online meeting. Sarah picks an available slot and sends the interview invitation. Emma receives a notification on the S&C platform, confirming the interview time and including a link to join the online meeting.

Creating a Questionnaire for Candidate Assessment To streamline the interview process, TechVision Corp uses a structured questionnaire to gauge candidates' technical knowledge and problem-solving skills. Sarah logs into the S&C platform and navigates to the "Assessment Tools" section. Here, she finds options to create a new questionnaire or select pre-made templates relevant to her internship role. Since she wants to evaluate specific skills related to TechVision's project requirements, Sarah opts to create a new questionnaire from scratch. She decides that the questionnaire should cover three main areas: technical knowledge (multiple-choice), coding skills (coding challenges), and scenario-based problem-solving (short-answer questions). After building the questionnaire, Sarah reviews each question for clarity, relevance, and alignment with the internship's objectives. She configures the scoring criteria for each section. After the questionnaire has been completed, she send and publish this questionnaire to all the applicants applied for the internship position.

Answering to questionnaire Emma logs into her S&C account and sees a notification on her dashboard about a new task. She clicks on the alert, which opens a detailed view of her application progress. A message from TechVision Corp indicates that she needs to complete a pre-interview questionnaire within three days. Emma also receives an email reminder with a direct link to the questionnaire. Emma clicks on the link in her dashboard, which takes her to the "Assessment" section of her application profile. Here, she sees an overview of the questionnaire, including the expected completion time. After completing all sections, the platform allows Emma to review her responses. She takes a final look to confirm that she hasn't missed any questions and feels satisfied with her answers. Emma clicks "Submit," and the S&C platform displays a confirmation message, assuring her that her responses have been successfully sent to TechVision Corp's HR team.

Finalizing the Evaluation and Making a Decision After interviewing multiple candidates, Sarah accesses the S&C platform to review her evaluation notes, assessment scores, and feedback from each candidate's questionnaire. After finalizing her decision, Sarah updates Emma's application status to "Accepted" on the S&C platform. This change automatically notifies Emma, informing her that she has been selected for the internship and can review her offer details.

University Internship Monitoring The university representative, Professor Green, logs into the

S&C platform. Upon accessing the university dashboard, she sees an overview of all ongoing internships involving students from her university. Professor Green selects the "Internship Monitoring" tab, where she views a list of students currently interning, along with detailed internship information such as company names, role titles, start and end dates, and supervisors. Each entry shows the student's progress and any recorded evaluations or feedback from the company.

Student Complaint Submission Mia, a student currently in an internship, encounters issues regarding her working hours. She logs into her S&C profile, navigates to "Internship Support," and selects "Submit Complaint.". Mia fills in the complaint form, specifying the issue and selecting "Excessive Work Hours" from a list of predefined categories. She adds further comments to explain her situation and attaches relevant evidence, such as email correspondence with her university supervisor.

Receiving Complaint By University Professor Green, who is responsible for overseeing student internships at the university, wants to check if any students have submitted complaints regarding their internships. Using the S&C platform, she can efficiently access and manage all student complaints in one place. Professor Green logs into the S&C platform and navigates to the "Complaints" section in her dashboard. Here, she has a dedicated view displaying all submitted complaints from students at her university. On this page, she sees a list of complaints categorized by student names, company names, submission dates, and complaint categories (e.g., "Work Hours," "Workplace Environment,"). Each entry also has a brief description of the issue for quick reference. To prioritize her review, Professor Green uses filtering options to sort complaints by severity, date submitted, or status (e.g., "New," "In Progress," "Resolved"). This helps her quickly identify urgent issues.

2.1.2 Class Diagram

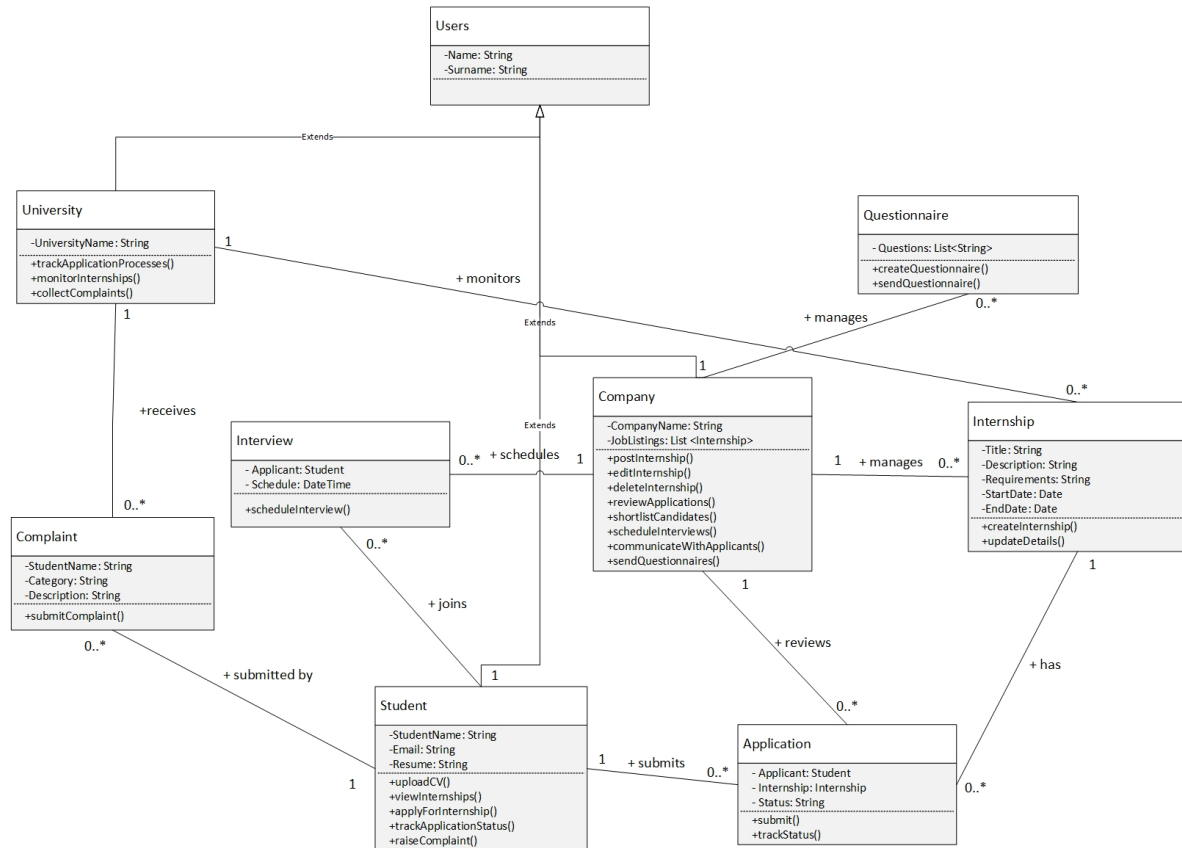


Figure 1: Class Diagram

2.1.3 State Diagram

Login. If an already registered user wants to login into their S&C account, they must provide the login credentials which includes email and password used in the time of the registration. If the credentials are accurate-matching those of the registered user. S&C shows the user's homepage, showcasing Internships, and on going Applications in which the user participating in. On the other hand, if the credentials are invalid up to third try, the login will be denied.

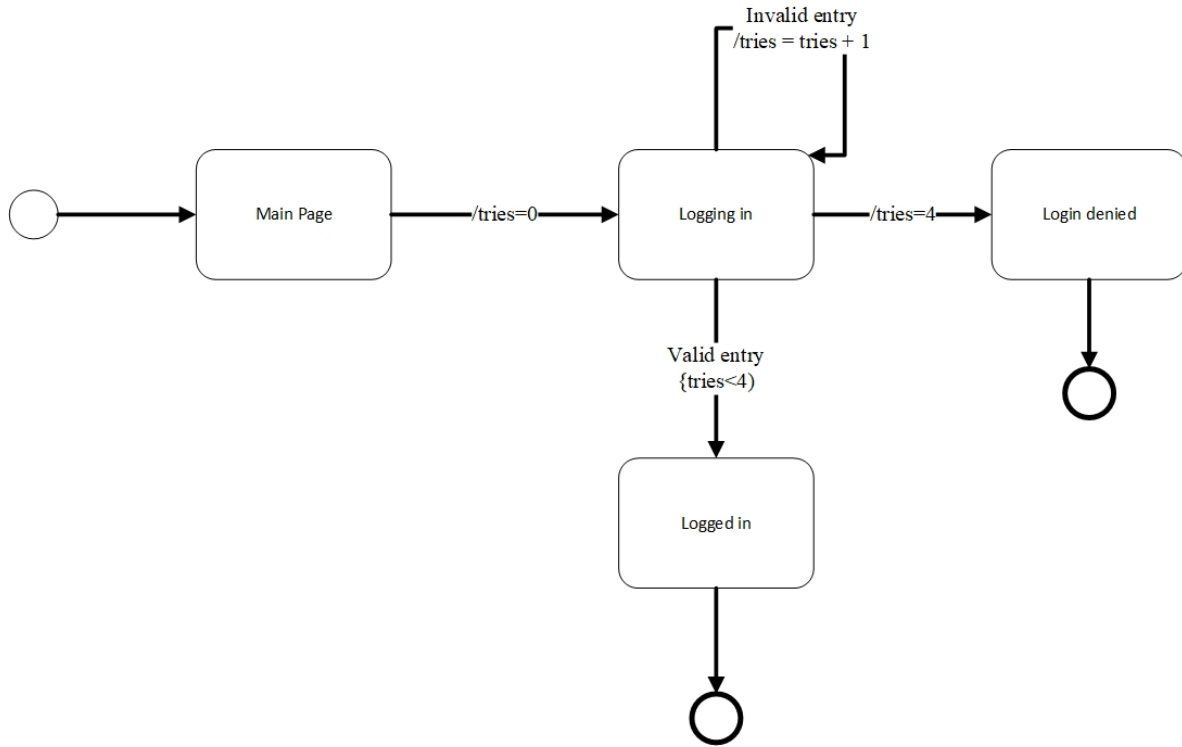


Figure 2: Login State Chart

Posting internships. When a Company User wants to post an internship, he/she is required to fill a form with specific details about the internship in the form, including job title, requirements, location, and date and application deadline or any specific rules associated with this position. If any of the entered parameters/details are known invalid by the system, S&C shows an error message, and redirect the user back to the post internship page. If all the entered parameters/details meet the system's criteria, then, S&C create the internship posting. The new created internship is then added to the list of internships.

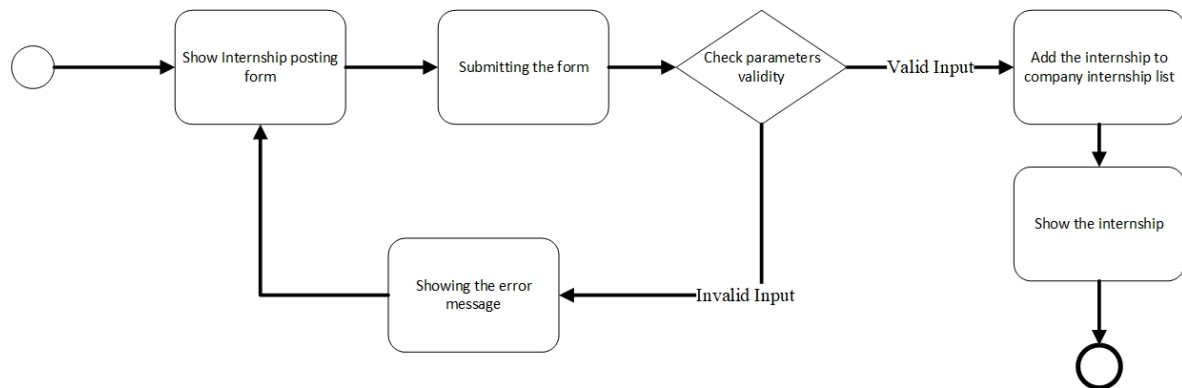


Figure 3: Internship Posting State Chart

View internship listing. With the use of the search bar user can view internships that they are looking for, just by entering the name or a keyword with respect to the details they want to find. When the parameter is entered, S&C will look for the internships that contain the entered parameter. Then the search results are shown to the user. Then user can click on specific internship from the search results, and S&C redirect user to the internship dashboard for detailed information and interaction with the internship.



Figure 4: Search and View Internships State Chart

Send application by a student. Once the student found the specific internship which matches for her/his interest and skills. Student has an option to apply and send his/her resume. S&C first performs a check to valid the application deadline of the internship. If student is applied throughout the allowed time-frame (before the deadline), S&C adds the students resume and details to the list of all the applicants applied for that internship, and directs student to applied internship page in their dashboard. On the other hand, if student is trying the send the application after the application deadline, S&C shows an error message to the student inform them about the closed application portal and expiration, and redirects student to the internship listing page for finding another internship.

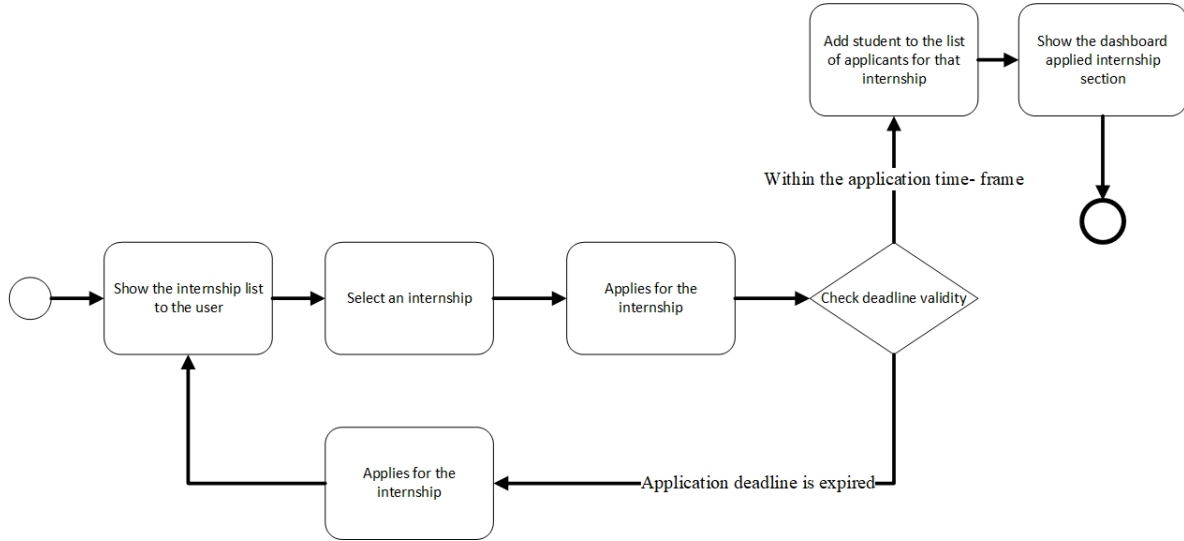


Figure 5: Sending Application for an Internships State Chart

View and shortlist the applicants for an internship by a company. A company user can goes to the internships listing page in the his/her dashboard, the user see all posted internship, and then, by selecting an internship, the user can see all the applicants applied for that position, and by clicking on a specific applicant user will redirect to a page that shows all the information about the applicants including resume. Furthermore, there is a search bar, by which the user can search specific keyword in the applicants entered information. Then, S&C shows the list of applicants matching the search keyword entered by the user. Then, user can short list the applicants after reviewing their qualifications.

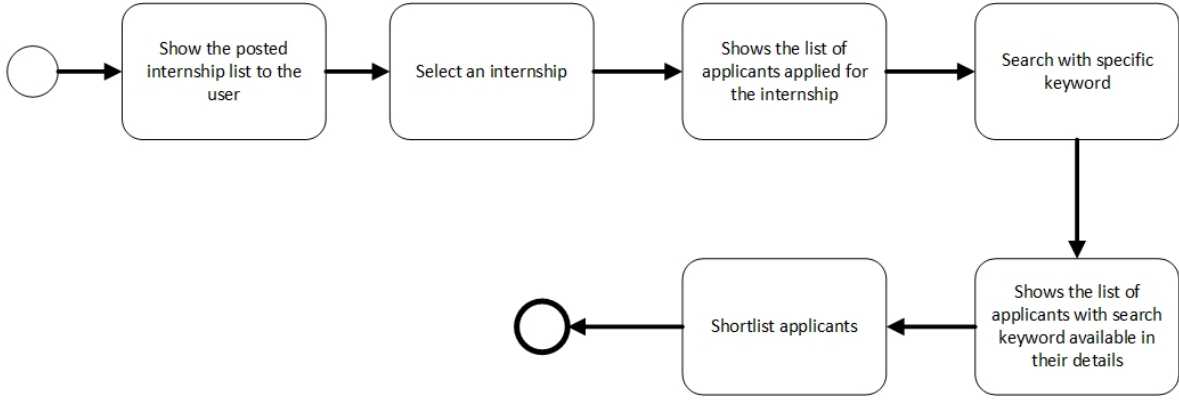


Figure 6: View and Shortlisting Applicants for an Internship State Chart

Scheduling an interview by a company. The process of scheduling an interview begins with a company representative logging into the platform and accessing their dashboard, where they can review the list shortlisted of applicants for a specific internship. After selecting an applicant for an interview, the representative navigates to the scheduling section and chooses a suitable date. Then, the S&C system checks the availability of the applicant in the selected date and time, if the applicant is not available, the S&C system displays an error message informing the company user about the availability, and company user should try with another time. On the other hand, if the applicant is available in that time, S&C system sends a meeting proposal to the applicant, if the applicant is confirmed the interview, S&C system will set the meeting date and time in the applicant's calendar. However, if the applicant does not accept the interview, the applicant will be rejected.

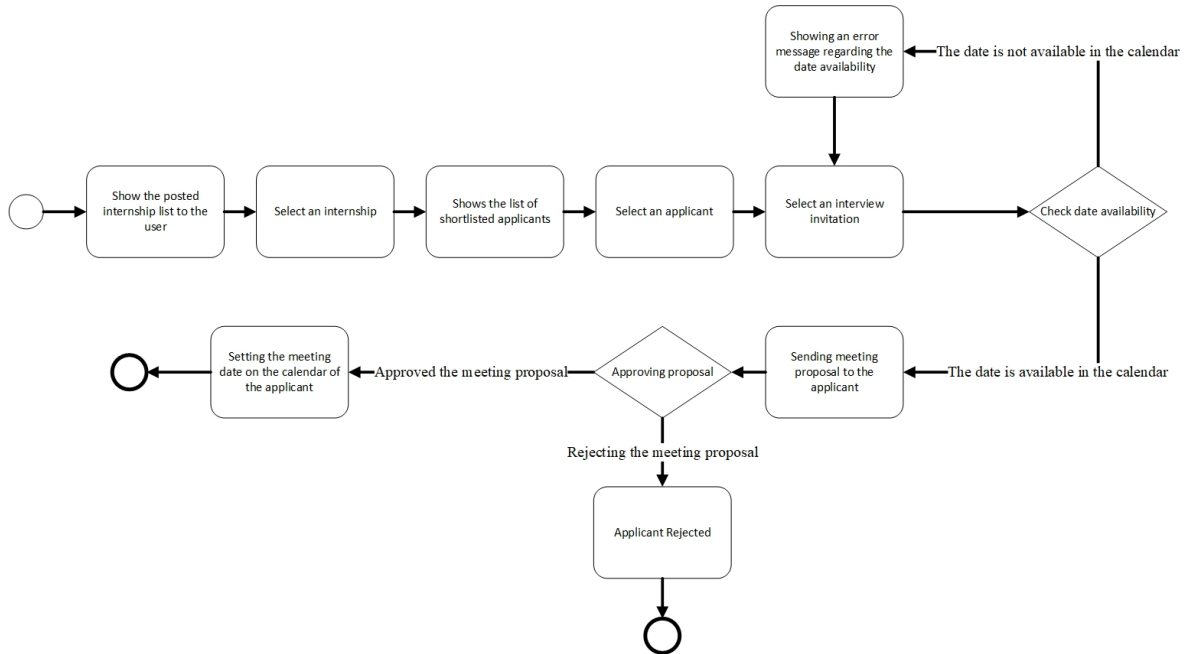


Figure 7: Scheduling Interview by a Company State Chart

Creating a questionnaire by a company. The process of creating a questionnaire for an internship begins by an already logged in company user, goes to section of posted internships, and by selecting an internship, company user navigate to creating a questionnaire form, and complete the form with question and details that the user wants to know about the applicants. After, the completion, the company user submits the form, the S&C system will check that if all the entered inputs are valid with respect to the criteria of the system, if the they were valid, the S&C system adds the questionnaire to the internship posting, and then sends notification to all the candidate that applied for that internship

about questionnaire. However, the inputs are invalid with respect to the previous defined structure in the S&C system, the S&C system will display an error message to the company user, the it redirects the user to the posted internship page.

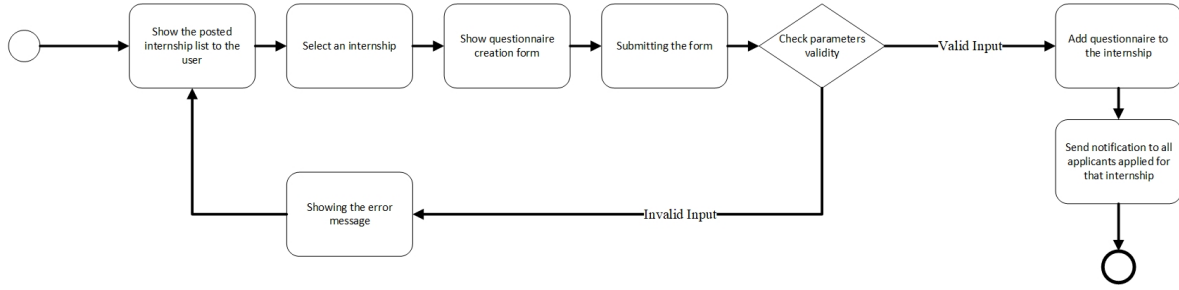


Figure 8: Creating a Questionnaire by a Company State Chart

Answering the questionnaire by a student. After that the company, create a questionnaire for an internship that they posted, the notification is sent to all the applicants applied for that internship. Then, the student sees the notification, and considering that he is already logged in to the S&C platform, the student user goes to the application section, in which the user sees all the applications that is sent for internships. Then, by selecting the the specific internship application, he/she will be go the detail page of the application, in which the user can see the application, and the new questionnaire associated with that application. Then, the user navigates to the option of filling the questionnaire. If this form has already been completed by the user, the S&C system shows an error message to the user and notify the user that the user about the form has already been filled. On the other hand if this is the first time, the user will be redirected to questionnaire page, the user then reviews all the instructions associated with this assessment, and then the user fills the form and submits. Following this, the S&C system will check the validity of the inputs, if all the inputs are valid the questionnaire will be submitted, and the user will be redirected to application page. However, if the input is invalid, the S&C shows the error message to the user.

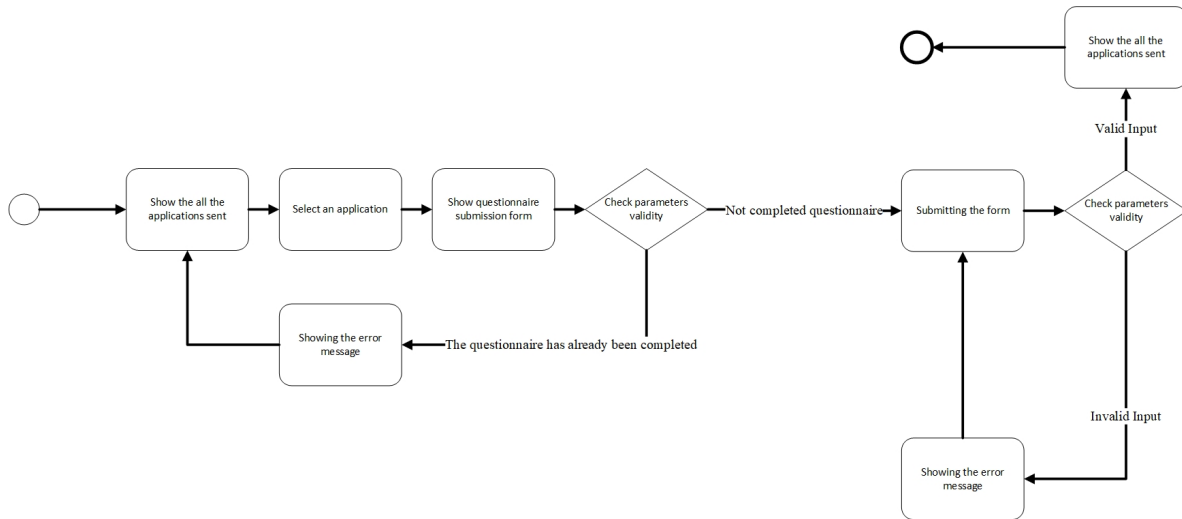


Figure 9: Answering a Questionnaire by a Student State Chart

Making the final decision about an internship by a company. Considering an already logged in company user wants to make final decision about an internship. The user goes on posted internship page, and then the user selects the internship that have a final decision. Then, the user view all the shortlisted applicants for that internship. Then, the user selects an applicant for the offer. First, the S&C system checks the availability of the applicants regarding the internship period. If the applicant is not available or he/she has another internship in progress during that period, the

applicant will be rejected, and the S&C shows an error message informing the company user about the situation, and then S&C redirects the company user to the shortlist page of the applicants. However, if the applicant is available, the company user sends the internship offer to the applicant, and awaits the applicant response. If the applicant accept the offer, the system will automatically change the status of all other applicant to rejected and the applicant of interest to accepted, and finally the system sends notification to all the applicants. However, if the applicant reject the internship offer, the system update the status of applicant and send the notification of rejection to that applicant.

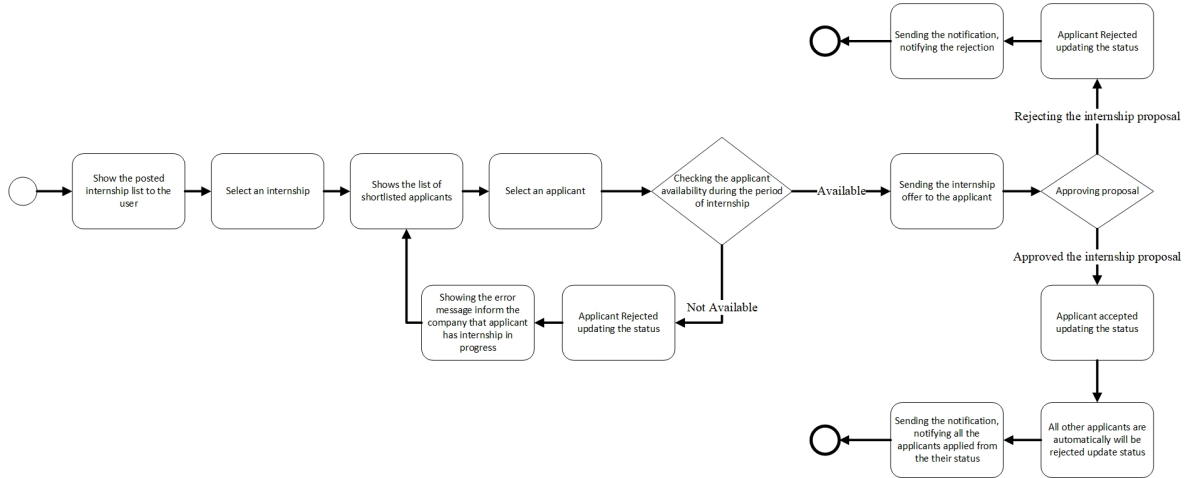


Figure 10: Making a Final Decision by a Company State Chart

University monitors an internship. Considering the university representative user is already registered and logged in to the websites. The user then goes to the on-going internship page, which represents all the internships that involve their students (this is identified during the registration time when the students insert the university affiliation). By selecting the an internship, they can see all the detail about the progress of the student that is taking the internship.

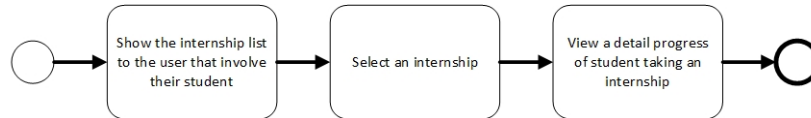


Figure 11: Monitoring Internship by a University State Chart

Student submits a complaint for an internship to university. Considering an already registered and logged in student, with correct affiliation of the university. Student user goes on the page of on-going internship, in which he sees all the internship that user involved such as, applied for, rejected, accepted or in-progress. The student user then selects the internship that the user wants to submits a complaint for. Then, by accessing he complaint form in the internship page, the student clicks on filling the form. The S&C system first check if the student is somehow involves in the internship or not (based on the status that described above). If the student is involved in the internship, he/she can submit the form. Then the S&C checks that the input is valid, if the input is valid, the form will be submitted, and the status of complaint will be set to submitted, and the S&C sends notification to the representative university of the student. However, if the input entered by the student is invalid, the S&C shows an error message to the student, and informing the user about the wrong input, and redirects the student to the list of on-going internship page. Furthermore, if the student is not involved in the internship by all means, the S&C system will return an error.

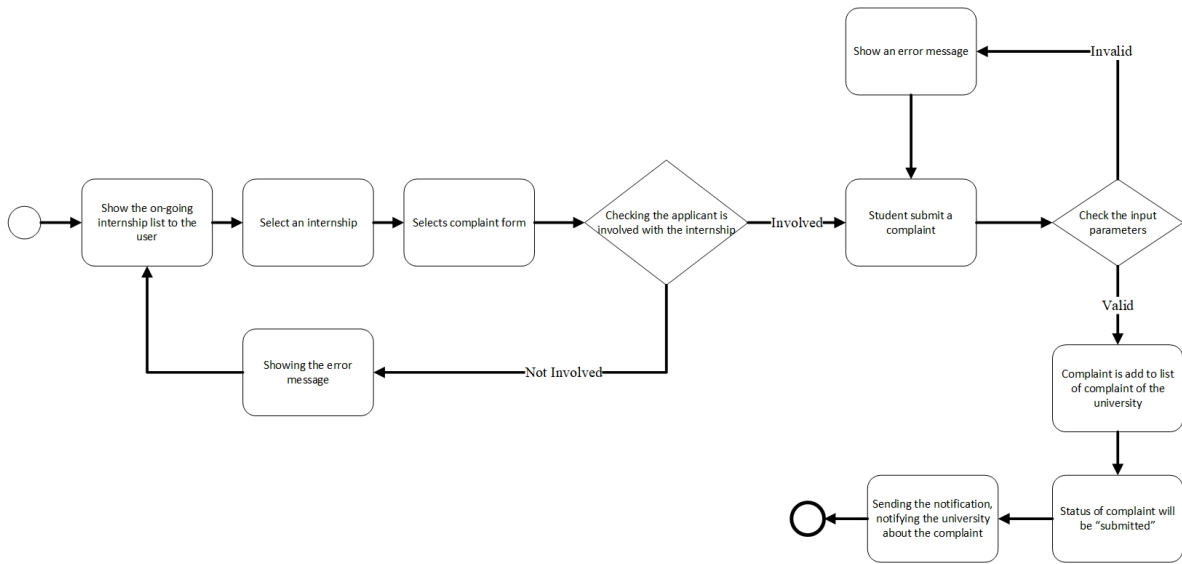


Figure 12: Submitting an Complaint by a Student State Chart

Receiving complaint by the university. Considering the university representative user is already registered and logged in to the websites. The user then goes to the on-going internship page, which represents all the internships that involve their students (this is identified during the registration time when the students insert the university affiliation). By selecting the an internship, they can see all the detail, and all the complaint submitted by their student to that internship. By clicking on the complaints, the user goes to page of complaints of that internship, then, the university user selects an specific complaint. After selecting, the S&C system changes the status of complaint to "In-Progress".

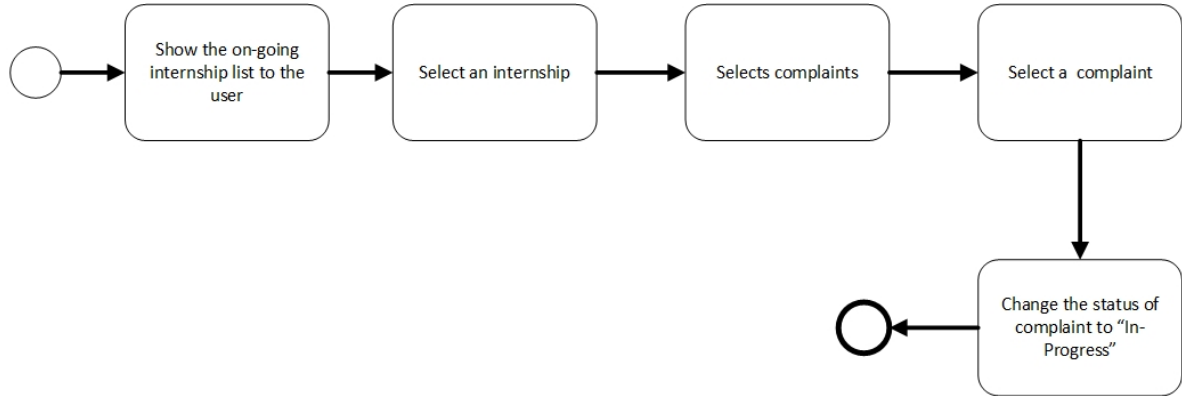


Figure 13: Receiving Complaint by a University State Chart

Logout. The users can logout from their account at anytime by clicking the logout button. After clicking the logout button, the S&C system will terminate the user session, and ensures a secure logout. Then, the user redirected to the login page, providing them with the opportunity to rejoin if they wish too.

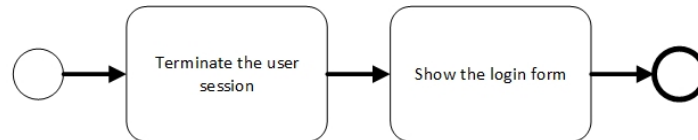


Figure 14: User Logout State Chart

2.2 Product Functions

Here are described the majority of functions of the systems.

Sign-up: The user enters name, last name, email address, and the university affiliation and a new password. Then an confirmation mail is required to complete the process

Login: The User sing in after type his/her email address and the password used in the time of registration in the login form.

Posting internship: The company user creates an new internship posting, by providing the information including job title, requirements, location, and date and application deadline or any specific rules associated with this position. The internship is then added to the list of internships in the system.

View internship listing: The student user can view all the available internships, and search through the system by entering keywords about an internship in the search bar. and the he/she can select an internship by selecting "more details", to see the page of internship.

Applying for an internship: The student user can select an internship, and clicks an bottom "Apply" in the internship page, and send his/her resume for the internship position. An student can send an already uploaded resume in the system, or select another resume for the company.

View applicants for an internship: The company user can view all the applicants applied for their internship posting, by going on the page of internship posting, and selecting an specific internship by selecting "More details", then going to internship page and clicking the bottom "Applicants", in which the can see all the applicants.

Shortlisting applicants for an internship: The company user can view all the applicants applied for their internship posting, by going on the page of internship posting, and selecting an specific internship by selecting "More details", then going to internship page and clicking the bottom "Applicants", in which the can see all the applicants. After this company user can view all the information including personal details and resume of applicants by selecting them. After review all the applicant, the company user can shortlist applicant by clicking the bottom "shortlist" on the applicant page.

Scheduling an interview: The company user can view all the applicants applied for their internship posting, by going on the page of internship posting, and selecting an specific internship by selecting "More details", then going to internship page and clicking the bottom "Applicants". Then by clicking the bottom "shortlisted applicants", the company user can see all the shortlisted applicants and then by going to applicant page. Then select "Scheduling an Interview" bottom and choose a suitable date and time for an interview.

Creating a questionnaire: The company user can create an questionnaire for an internship posting, by going to the page of internship posting of the company, and then selecting the internship by clicking the bottom "More Details", then clicking the bottom "Creating Questionnaire", and then creates a questionnaire for all the applicants applied for that internship. When a new questionnaire is created S&C sends notification to all the applicants applied for that position.

Answering the questionnaire: The student user can answer the questionnaire of an specific internship, by going to page of applications. Then selecting the specific application, and then going to page of application, and the student user can answer the questionnaire regarding that internship.

Decision on applications: The company user can update the status of applicants based on the evaluation on applicants resume, interview, and questionnaire assessment and can accept, and reject an applicant for a position.

Monitoring internships: The university users can monitor the internships of all of their students that are involving with an internship, and review their progress and status about the internship.

Submission of complaints: The student users can send complaints regarding the internships that they are involved with to their university.

Receiving complaints: The university users can view all the complaints of their students involved with an internship, in the on-going internship page.

2.3 User Characteristics

There are three types of registered users in the S&C system: Company Users, Student Users, and University Users. Here is the brief explanation of their main characteristics:

Company Users: Company users join S&C platform in order to recruit students for the internship offering of their company. They need to have a device with stable internet connection and an account in order to access the S&C platform, in which they can manage internship posting (create, edit, delete), review applications, scheduling interview, creating questionnaire for applicant assessments, and selecting suitable applicants for the internship positions.

Student Users: Student users join S&C platform in order to find a suitable internship positions offering from various companies. They need to have a device with stable internet connection and an account in order to access the S&C platform, in which, they can manage their resume (upload, edit, delete), look for internship by searching through the system, review internship details and requirements, applying for an internship, take part in interview scheduled by a company, take an assessment tests provided by the company, tracking the status of their application, and making complaints to the university about the internships they are involved in.

University Users: University users join S&C platform in order to track internships and manage complaints of their students. They need to have a device with stable internet connect and an account in order to access the S&C platform, in which, they can track internships involved their students, and receive complaints from their students involving internships.

2.4 Assumptions, Dependencies and Constraints

2.4.1 Domain Assumptions

Table 5: Domain Assumptions

ID	Description
DA1	Student users have a valid email address.
DA2	Company users have a valid email address.
DA3	University users have a valid email address.
DA4	Student users need to have device with stable internet connection.
DA5	Company users need to have device with stable internet connection.
DA6	University users need to have device with stable internet connection.
DA7	All users need to enter the correct information in the registration phase
DA8	Company users need to enter the correct information in the internship posting
DA9	Student users need to enter the correct information in the resume

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The S&C website is designed for both students and companies and universities and is accessible to anybody with an Internet browser and a steady connection.

3.1.2 Hardware Interfaces

The system will be available from any device having an Internet Browser and a stable Internet connection. The user can utilize any device, including a PC, tablet, or smartphone. However, using a PC is recommended.

3.1.3 Software Interfaces

The system requires the Zoom/Teams APIs to enable real-time video and audio communication. These APIs are essential for holding meetings including meeting scheduling, joining, and session control. Also, it will use a mailing system to send eMails to the Users during the registration process, application process. The system requires integration with map APIs (such as Google Maps or OpenStreetMap) to handle location-based functionalities for internship posting. The system utilizes Calendar APIs (such as Google Calendar API or Microsoft Outlook Calendar API) to manage scheduling and time-based functionalities including meetings, internships deadline and more.

3.1.4 Communication Interfaces

The communication Interfaces needed by the system are the HTTPS Protocol and the Mail System Transfer Protocol (SMTP).

3.2 Functional Requirements

Table 6: Requirements

Requirement ID	Description
R1	S&C allows unregistered end users to register.
R2	After an end user fills all the blanks in registration page correctly, the system must send an email to her/him, in order to confirm her/his email.
R3	S&C allows logged out Student users to login.
R4	S&C allows logged out Company users to login.
R5	S&C allows logged out University users to login.
R6	Users need to have device with stable internet connection.
R7	S&C allows Student users to upload their resume.
R8	S&C allows Student users to edit their resume.
R9	S&C allows Student users to delete their resume.
R10	S&C allows Student users to search for internship posting.
R11	S&C allows Student users to view a list internship postings.
R12	S&C allows Student users to send application (resume) for an internship posting.
R13	S&C allows Student users to track the status of their applications.
R14	S&C allows Student users to view a list of their applications.
R15	S&C allows Student users to submit complaints about internship they are involved in to their university.
R16	S&C allows Student users to submit questionnaire assessment test for an internship.
R17	S&C allows Student users to take part in the meetings.
R18	S&C allows Student users to accept/reject internship offerings.
R19	S&C allows Student users to accept/reject meeting offerings.
R20	S&C allows Company users to enter information for internship offerings.
R21	S&C allows Company users to post/publish internship offerings.
R22	S&C allows Company users to edit posted internship offerings.
R23	S&C allows Company users to delete posted internship offerings.

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Requirement ID	Description
R24	S&C allows Company users to view all the applicant's application for an posted internship offering.
R25	S&C allows Company users to shortlist applicants for an posted internship offering.
R26	S&C allows Company users to send meeting proposal to shortlisted applicant for an posted internship offering.
R27	S&C allows Company users to create questionnaire for an posted internship offering.
R28	S&C allows Company users to send questionnaire for an posted internship offering.
R29	S&C allows Company users to edit questionnaire for an posted internship offering.
R30	S&C allows Company users to delete questionnaire for an posted internship offering.
R31	S&C allows Company users to send proposal/offer for an posted internship offering.
R32	S&C allows University users to track applications for an posted internship offering involved their students.
R33	S&C allows University users to receive complaints for an posted internship offering involved their students.
R34	S&C allows end users to view their information.
R35	S&C must notify Student users to about the submission of application.
R36	S&C must notify Student users to about the change of application's status.
R37	S&C must notify Student users to about the request for meeting from an applied internship.
R38	S&C must notify Company users to about the submission of questionnaire to an applied internship.
R39	S&C allows end users to edit their information.
R40	S&C stores the information about the Users.
R41	S&C shall ensure security of data.
R42	S&C must notify Company users to about the acceptance/rejection of request for meeting from an applicant.
R43	S&C must notify Company users to about the acceptance/rejection of the offering internship proposal from an applicant.
R44	S&C must notify University users to about the receiving complaints from students involved in internship.
R45	S&C must notify University users to about the change of status of applications from students involved in internship.

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Requirement ID	Description
R46	S&C must update the status of Student users application once a new change has been placed.
R47	S&C must update the calendar of Student users for meetings.
R48	S&C must update the calendar of Student users for internship periods.
R49	S&C must update the calendar of Company users for meetings.
R50	S&C must communicate with the mailing system in order to allow Users to register their account.
R51	S&C allows University users to search for and view a list of their students.
R52	S&C allows University users to search for and view a list of complaints from their students.
R53	S&C allows University users to search for and monitor internship involved with their students.
R54	S&C must notify University users to about the change of status of applications from students involved in internship.

3.2.1 Use Case Diagram

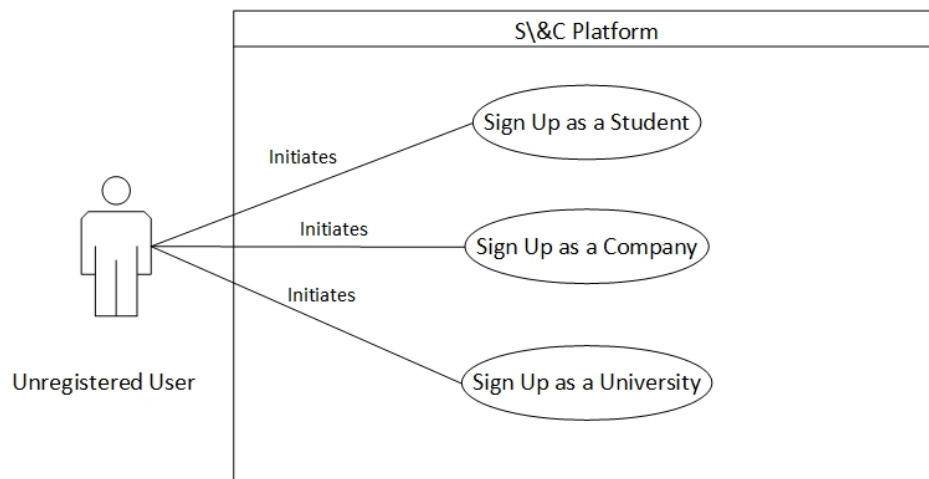


Figure 15: Use Cases Diagram for Unregistered Users



Figure 16: Use Cases Diagram for Students

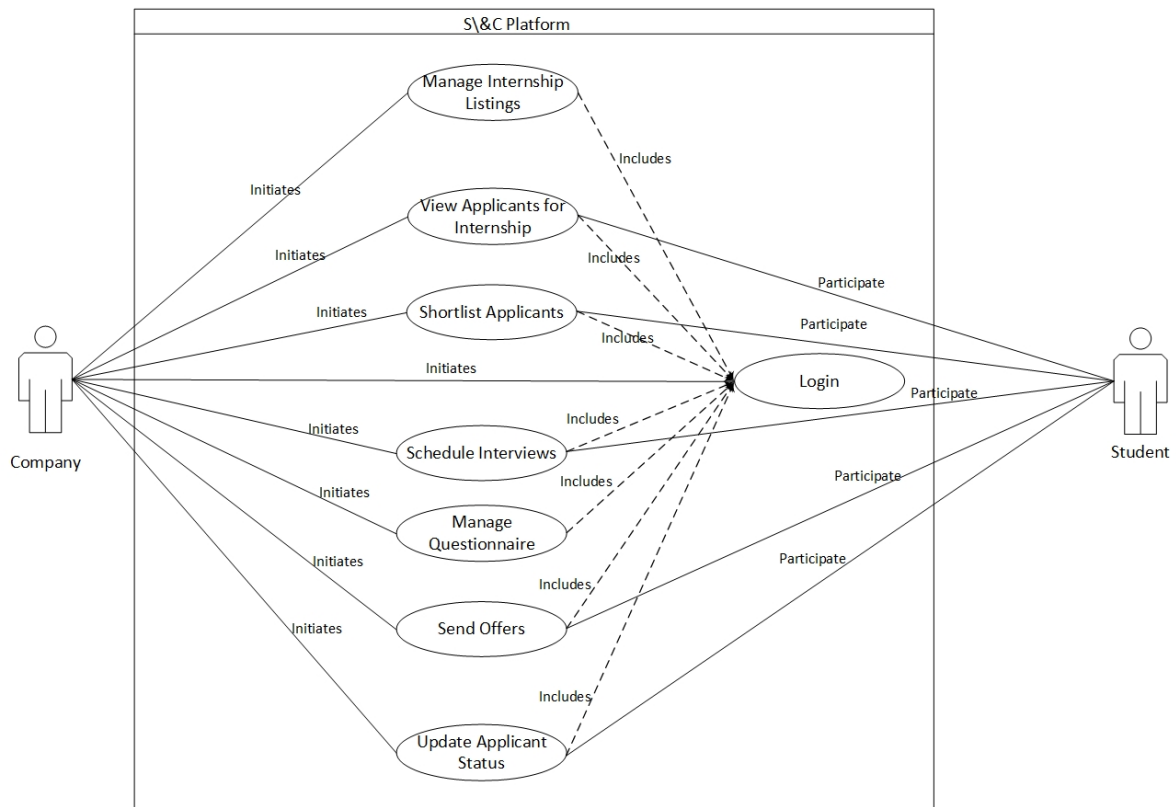


Figure 17: Use Cases Diagram for Company

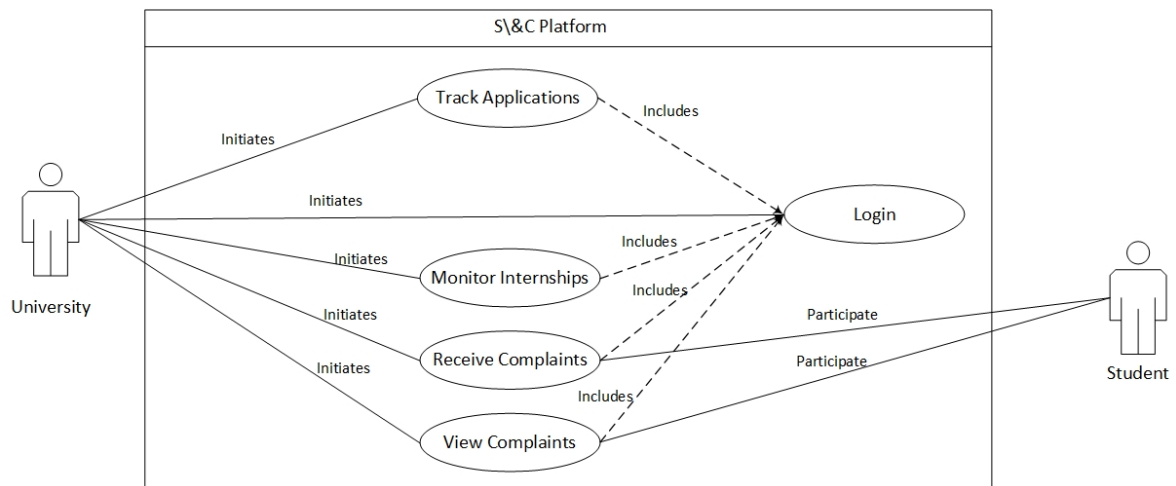


Figure 18: Use Cases Diagram for University

3.2.2 Use Cases

This section explains and represents the key identified use cases. The table includes entry conditions, event flow, exit conditions, and exceptions for each entity, as well as a sequence diagram depicting message exchange between entities and called functions.

Register as student

Table 7: Register as Student

Name	Register as student
Actor	Student
Entry Condition	Student user opens the application
Events Flow	1. S&C shows the login form. 2. The Student user clicks on "Create an Account" button. 3. S&C shows the sign-up form 4. The Student user inserts his name, surname, nickname, university affiliation, Email and password in the form and also ticks on the "Sign-up as Student" checkbox. 5. The Student user clicks on the "Register" button. 6. S&C checks all the credentials. 7. If credentials are correct S&C sends a confirmation email to the student through the email provider. 8. The Student user clicks on the confirmation link.
Exit Conditions	The student user is registered in the app.
Exceptions	1. The email address is already linked to an account. In this case an error message is shown and the Student user is redirected to the profile creation settings. 2. The entered password does not satisfy the password requirements. In this case an error message is shown and the Student user about weakness of password. 3. Student user doesn't check the policies checkbox. 4. Student user doesn't fill all the fields. In this case, application shows warning.

Register as company

Table 8: Register as Company

Name	Register as company
Actor	Company
Entry Condition	Company user opens the application
Events Flow	1. S&C shows the login form. 2. The Company user clicks on "Create an Account" button. 3. S&C shows the sign-up form 4. The Company user inserts his organization name, organization department, representative name, representative surname, representative position, company Email and password in the form and also ticks on the "Sign-up as Company" checkbox. 5. The Company user clicks on the "Register" button. 6. S&C checks all the credentials. 7. If credentials are correct S&C sends a confirmation email to the company through the email provider. 8. The company user clicks on the confirmation link.
Exit Conditions	The company user is registered in the app.
Exceptions	1. The email address is already linked to an account. In this case an error message is shown and the company user is redirected to the prole creation settings. 2. The entered password does not satisfy the password requirements. In this case an error message is shown and the Student user about weakness of password. 3. Company user doesn't check the policies checkbox. 4. Company user doesn't fill all the fields. In this case, application shows warning.

Register as university

Table 9: Register as University

Name	Register as university
Actor	University
Entry Condition	University user opens the application
Events Flow	1. S&C shows the login form. 2. The University user clicks on "Create an Account" button. 3. S&C shows the sign-up form 4. The University user inserts his university name, university department, representative name, representative surname, representative position, university Email and password in the form and also ticks on the "Sign-up as University" checkbox. 5. The university user clicks on the "Register" button. 6. S&C checks all the credentials. 7. If credentials are correct S&C sends a confirmation email to the university through the email provider. 8. The university user clicks on the confirmation link.
Exit Conditions	The university user is registered in the app.
Exceptions	1. The email address is already linked to an account. In this case an error message is shown and the university user is redirected to the prole creation settings. 2. The entered password does not satisfy the password requirements. In this case an error message is shown and the university user about weakness of password. 3. University user doesn't check the policies checkbox. 4. University user doesn't fill all the fields. In this case, application shows warning.

Login

Table 10: Login

Name	Login
Actor	End Users
Entry Condition	Entry Condition End user opens the application
Events Flow	1. S&C shows the login form. 2. The User inserts his email and password in the form. 3. The User clicks on the "Login" button. 4. S&C checks the credentials.
Exit Conditions	The system allows the user to login.
Exceptions	1. Incorrect email or password. An error message is shown and the User is redirected back to the Login page

Manage resume

Table 11: Manage Resume

Name	Manage resume
Actor	Student user
Entry Condition	1. End user open the application. 2. He/She already signed in.
Events Flow	1. The Student clicks on "Create a Resume" button. 2. S&C shows the resume form. 3. User selects an option to "Upload New Resume", "Edit Current Resume", or "Delete Current Resume" a resume. 4. S&C ask for user confirmation about the changes. 5. S&C processes the request. 6. S&C updates the profile with the changes.
Exit Conditions	The resume is uploaded, edited, or deleted, and the profile is updated.
Exceptions	1. Unsupported file format during upload. 2. Network interruption during the process.

Search internship

Table 12: Search Internship

Name	Search internship
Actor	Student user
Entry Condition	1. End user open the application. 2. He/She already signed in.
Events Flow	1. Student user clicks on the search bar. 2. Student user enters keywords or filters for internships. 3. The S&C system searches and displays relevant postings.
Exit Conditions	User views detailed information about searched internships.
Exceptions	1. No results matching the search criteria. In this case the User will be redirected back to the previous page. 2. System connectivity issues.

View internship info

Table 13: View internship info

Name	View internship info
Actor	Student user
Entry Condition	1. End user open the application. 2. He/She already signed in.
Events Flow	1. Student user clicks on the search bar. 2. Student user enters keywords or filters for internships. 3. The S&C system searches and displays relevant postings. 4. Student user clicks on an internship to view details.
Exit Conditions	User views detailed information about searched internships.
Exceptions	1. No results matching the search criteria. In this case the User will be redirected back to the previous page. 2. System connectivity issues.

Send application for internship

Table 14: Send application for internship

Name	Send application for internship
Actor	Student user, Company user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. He/She already has available resume. 4. He/She already selected an internship.
Events Flow	1. Student user clicks “Apply”. 2. Student user sends an already uploaded resume. 3. The S&C system verifies the application deadline. 4. The S&C system submits the application. 5. The S&C system notifies the company.
Exit Conditions	Application is submitted, and the student sees the “Applications” status.
Exceptions	1. Application submitted past the deadline. 2. Missing resume file.

Track application status

Table 15: Track application status

Name	Track application status
Actor	Student user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. He/She has already applied for internships.
Events Flow	1. Student user goes to dashboard. 2. Student user selects "Applications" option. 3. The S&C system displays the status of all submitted applications.
Exit Conditions	Student user views the status updates of their applications.
Exceptions	1. Application data retrieval error. 2. Student user does not have any applications

Submit complaint

Table 16: Submit complaint

Name	Submit complaint
Actor	Student user, University user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. He/She has already applied for internships.
Events Flow	1. Student user goes to dashboard. 2. Student user selects "Applications" option. 3. The S&C system displays all submitted applications. 4. Student user select an application of interest. 5. Student user accesses the "Application Support" section. 6. Student user fills out the complaint form and submits it. 7. The S&C system verifies the complaint. 8. The S&C system sends it to the affiliated university.
Exit Conditions	Complaint is submitted, and status changes to "Submitted."
Exceptions	1. Invalid or incomplete input in the complaint form. 2. Student user does not have any applications

Submit questionnaire

Table 17: Submit questionnaire

Name	Submit questionnaire
Actor	Student user, Company user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. He/She has already applied for internships. 4. He/She has been assigned a questionnaire.
Events Flow	1. Student user goes to dashboard. 2. Student user selects "Applications" option. 3. The S&C system displays all submitted applications. 4. Student user select an application of interest. 5. Student user accesses the questionnaire from their application dashboard. 6. Student user fills out the questionnaire form and submits it. 7. The S&C system verifies the questionnaire. 8. The S&C system sends it to the company.
Exit Conditions	Questionnaire is successfully submitted.
Exceptions	1. Attempt to submit after the deadline. 2. Invalid or incomplete input in the complaint form. 3. Student user does not have any applications

Accept/Reject internship offer

Table 18: Accept/Reject internship offer

Name	Accept/Reject internship offer
Actor	Student user, Company user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. He/She has already applied for internships. 4. He/She has received an internship offer.
Events Flow	1. Student user goes to dashboard. 2. Student user selects "Applications" option. 3. The S&C system displays all submitted applications. 4. Student user select an application of interest. 5. Student user reviews the offer in their application dashboard. 6. Student user chooses to accept or reject the offer. 7. The S&C system updates the status. 8. The S&C system notifies the company.
Exit Conditions	Offer is accepted or rejected, and the status is updated.
Exceptions	1. Student user does not have any applications. 2. Expired offer.

Attend scheduled meetings

Table 19: Attend scheduled meetings

Name	Attend scheduled meetings
Actor	Student user, Company user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. He/She has already applied for internships. 4. He/She has is notified of a scheduled meeting.
Events Flow	1. Student user goes to dashboard. 2. Student user selects "Applications" option. 3. The S&C system displays all submitted applications. 4. Student user select an application of interest. 5. Student user accesses the meeting link or schedule from their application dashboard. 6. The S&C system confirms attendance and records participation.
Exit Conditions	Meeting is attended and marked as complete.
Exceptions	1. Student user does not have any applications. 2. Expired date. 3. Connectivity issues during the meeting.

View application list

Table 20: View application list

Name	View application list
Actor	Student user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. He/She has already applied for internships.
Events Flow	1. Student user goes to dashboard. 2. Student user selects "Applications" option. 3. The S&C system displays all submitted applications.
Exit Conditions	Student user reviews the list of applications.
Exceptions	1. Student user does not have any applications. 2. The S&C system failure in retrieving application data.

Manage internship listings

Table 21: Manage internship listings

Name	Manage internship listings
Actor	Company user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The Company user is on his/her profile page.
Events Flow	1. Company user goes to dashboard. 2. Company user selects an option to "Create New Internship", "Edit Current Internship", or "Delete Current Internship" a resume. 3. The S&C system validates the input or processes the deletion.
Exit Conditions	Internship listings are updated, created, or deleted.
Exceptions	1. Invalid input during creation or editing. 2. Deletion of an already archived internship.

View applicants for internship

Table 22: View applicants for internship

Name	View applicants for internship
Actor	Company user, Student user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The Company user is on his/her profile page. 4. The Company user selects an internship listing.
Events Flow	1. Company user goes to dashboard. 2. Company user goes to posted internships. 3. Company user navigates to the "Applicants" section of the internship listing. 4. The S&C system displays a list of applicants and their details.
Exit Conditions	Company user views the list of applicants.
Exceptions	1. No applicants yet submitted applications. 2. The S&C error in retrieving applicant data.

Shortlist applicants

Table 23: Shortlist applicants

Name	Shortlist applicants
Actor	Company user, Student user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The Company user is on his/her profile page. 4. The Company user selects an internship listing.
Events Flow	1. Company user goes to dashboard. 2. Company user goes to posted internships. 3. Company user navigates to the "Applicants" section of the internship listing. 4. Company user reviews applications. 5. Company user selects applicants to shortlist. 6. The S&C system updates the shortlist status for selected applicants.
Exit Conditions	Shortlist is updated and saved.
Exceptions	1. No applicants yet submitted applications. 2. The S&C error while saving the shortlist.

Schedule interviews

Table 24: Schedule interviews

Name	Schedule interviews
Actor	Company user, Student user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The Company user is on his/her profile page. 4. The Company user selects an internship listing. 5. The Company user has shortlisted applicants for an internship.
Events Flow	1. Company user goes to dashboard. 2. Company user goes to posted internships. 3. Company user navigates to the "Applicants" section of the internship listing. 4. Company user selects an applicant in the shortlist. 5. Company user selects an applicant and a time slot for the interview. 6. The S&C system verifies the availability of both parties and schedules the meeting.
Exit Conditions	Interview is scheduled, and both parties are notified.
Exceptions	1. Conflict in availability. 2. Network issues during scheduling.

Manage questionnaire

Table 25: Manage questionnaire

Name	Manage questionnaire
Actor	Company user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The Company user is on his/her profile page. 4. The Company user selects an internship listing.
Events Flow	1. Company user goes to dashboard. 2. Company user goes to posted internships. 3. Company user chooses to create, edit, or delete a questionnaire. 4. The S&C system validates the inputs or processes the deletion. 5. The S&C system questionnaire information is saved or removed.
Exit Conditions	Questionnaire is updated, created, or deleted.
Exceptions	1. Invalid input during creation or editing. 2. Attempting to delete a questionnaire not associated with the internship.

Send offers

Table 26: Send offers

Name	Send offers
Actor	Company user, Student user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The Company user is on his/her profile page. 4. The Company user selects an internship listing. 5. The Company user has already evaluated candidates.
Events Flow	1. Company user goes to dashboard. 2. Company user goes to posted internships. 3. Company user navigates to the "Applicants" section of the internship listing. 4. Company user selects an applicant in the shortlist. 5. Company user selects an applicant and prepares an offer. 6. The S&C system sends the offer to the student.
Exit Conditions	Offer is sent, and the status is updated.
Exceptions	1. Error in offer details or transmission. 2. Student user has already involve with an internship.

Update applicant status

Table 27: Update applicant status

Name	Update applicant status
Actor	Company user, Student user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The Company user is on his/her profile page. 4. The Company user selects an internship listing. 5. The Company user has already evaluated candidates.
Events Flow	1. Company user goes to dashboard. 2. Company user goes to posted internships. 3. Company user navigates to the "Applicants" section of the internship listing. 4. Company user selects an applicant in the shortlist. 5. Company user updates the status of applicants (e.g., shortlisted, rejected). 6. The S&C system saves the changes and notifies the applicants.
Exit Conditions	Applicant statuses are updated.
Exceptions	1. System error in updating statuses.

Track applications

Table 28: Track applications

Name	Track applications
Actor	University user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The University user is on his/her profile page. 4. The University user selects "Applications".
Events Flow	1. University user goes to dashboard. 2. University user goes to "Applications". 3. University user navigates to the "Applicants". 4. University user selects an applicant. 5. The S&C system retrieves and displays relevant application data.
Exit Conditions	University user reviews application data.
Exceptions	1. No applications linked to students from their institution. 2. System error in retrieving data.

Monitor internships

Table 29: Monitor internships

Name	Monitor internships
Actor	University user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The University user is on his/her profile page. 4. The University user selects "On-going Internships".
Events Flow	1. University user goes to dashboard. 2. University user goes to "On-going Internships". 3. University user selects an internship. 4. The S&C system retrieves and displays student progress and feedback.
Exit Conditions	University User reviews the status and details of internships.
Exceptions	1. No active internships for their students. 2. System error in retrieving data.

Receive and view complaints

Table 30: Receive and view complaints

Name	Receive and view complaints
Actor	University user, Student user
Entry Condition	1. End user open the application. 2. He/She already signed in. 3. The University user is on his/her profile page.
Events Flow	1. University user goes to dashboard. 2. University user goes to "Complaints". 3. University user views a list of submitted complaints. 4. The S&C system categorizes and displays the details for review.
Exit Conditions	University user reviews the status and details of internships.
Exceptions	1. No complaints submitted. 2. Error in retrieving complaint data.

3.2.3 Sequence Diagram

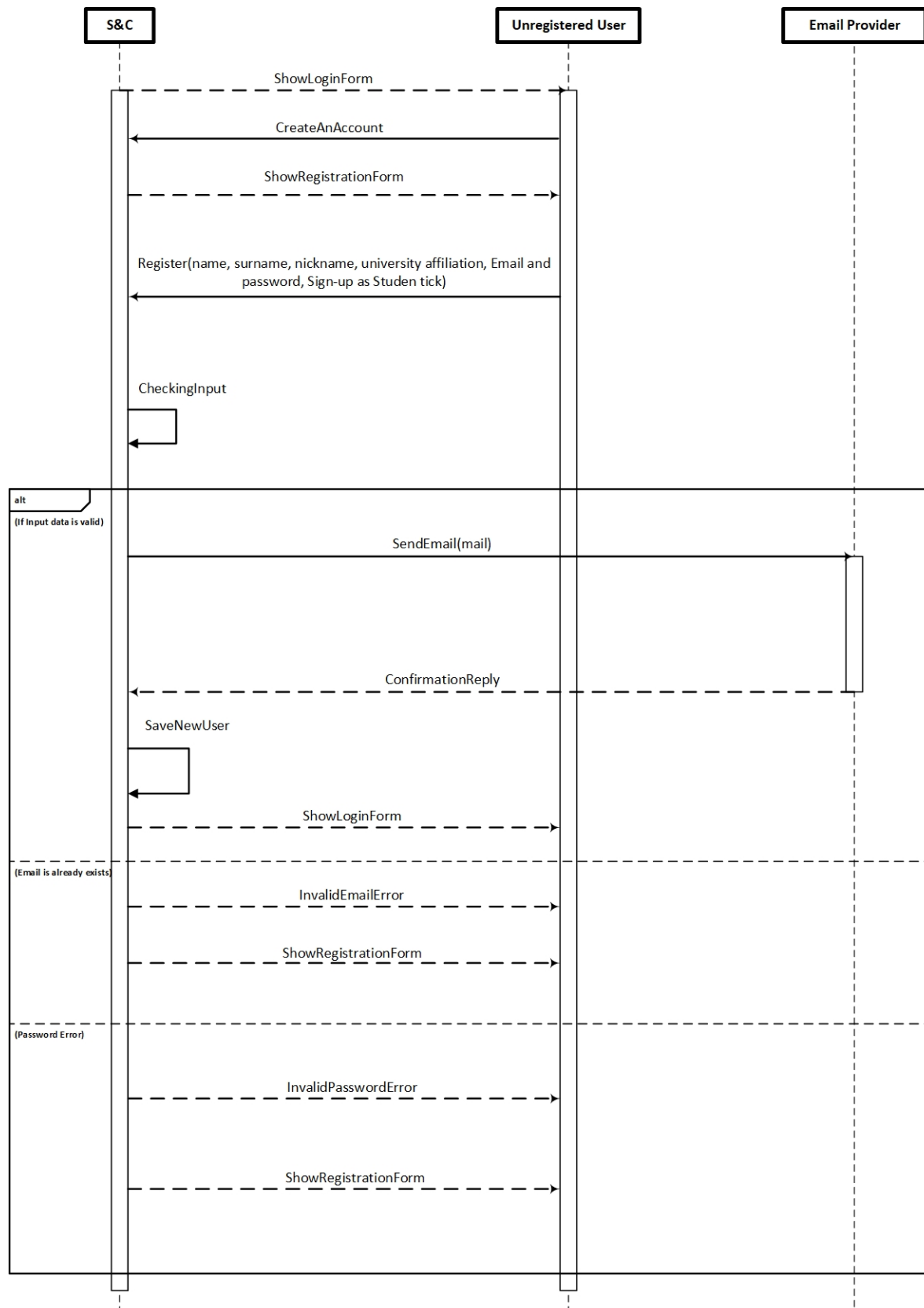


Figure 19: Registration as Student Sequence Diagram

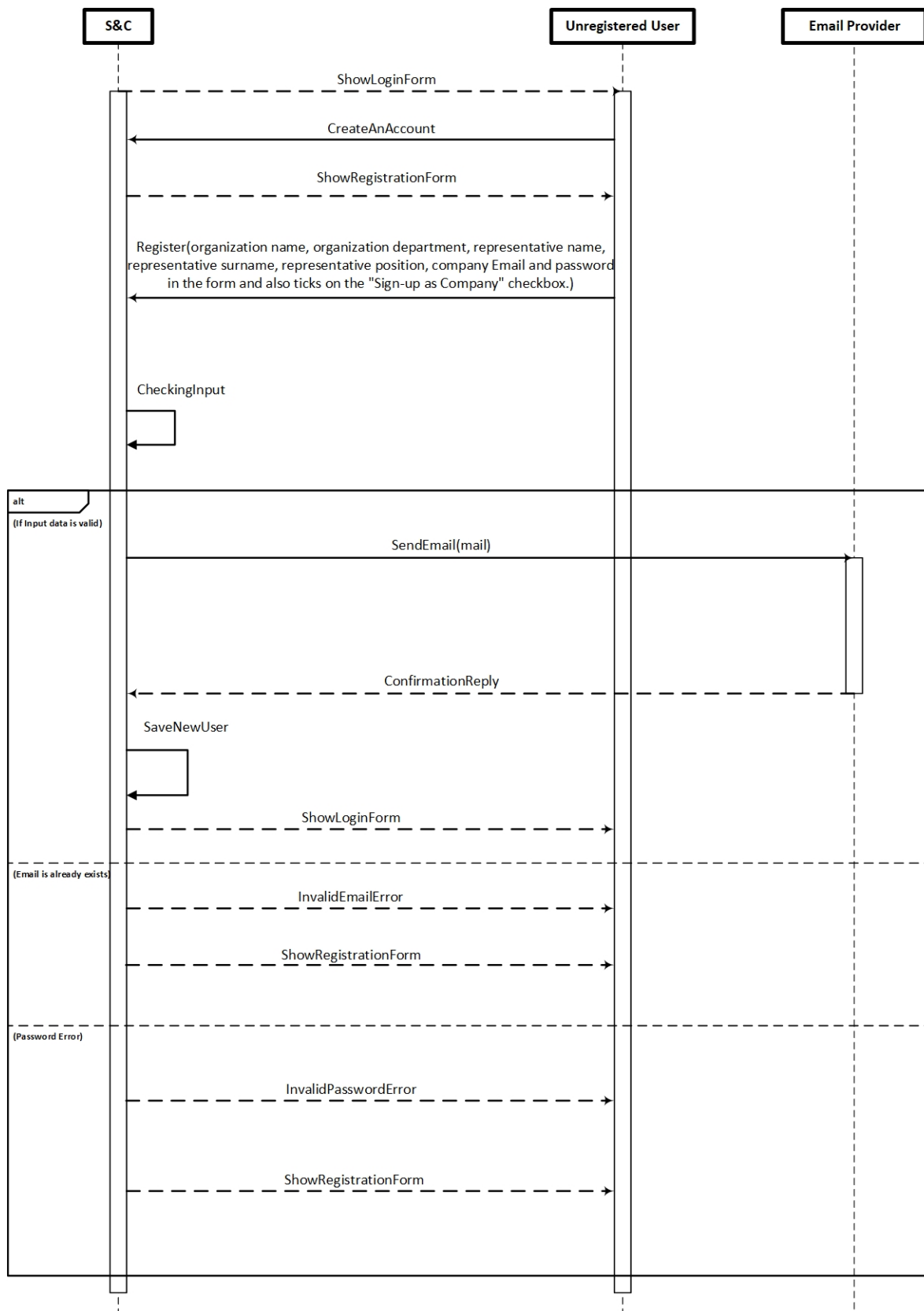


Figure 20: Registration as Company Sequence Diagram

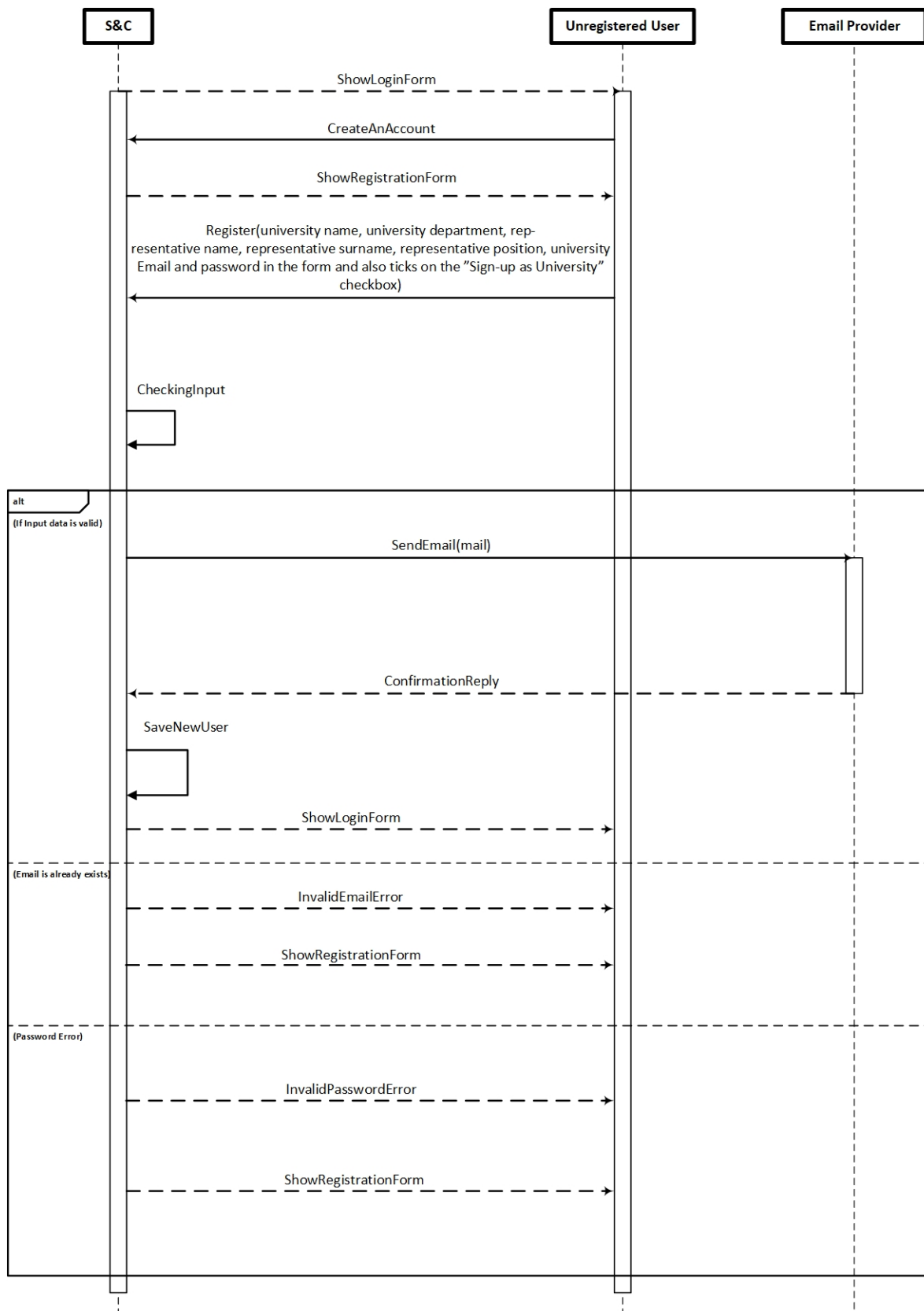


Figure 21: Registration as University Sequence Diagram

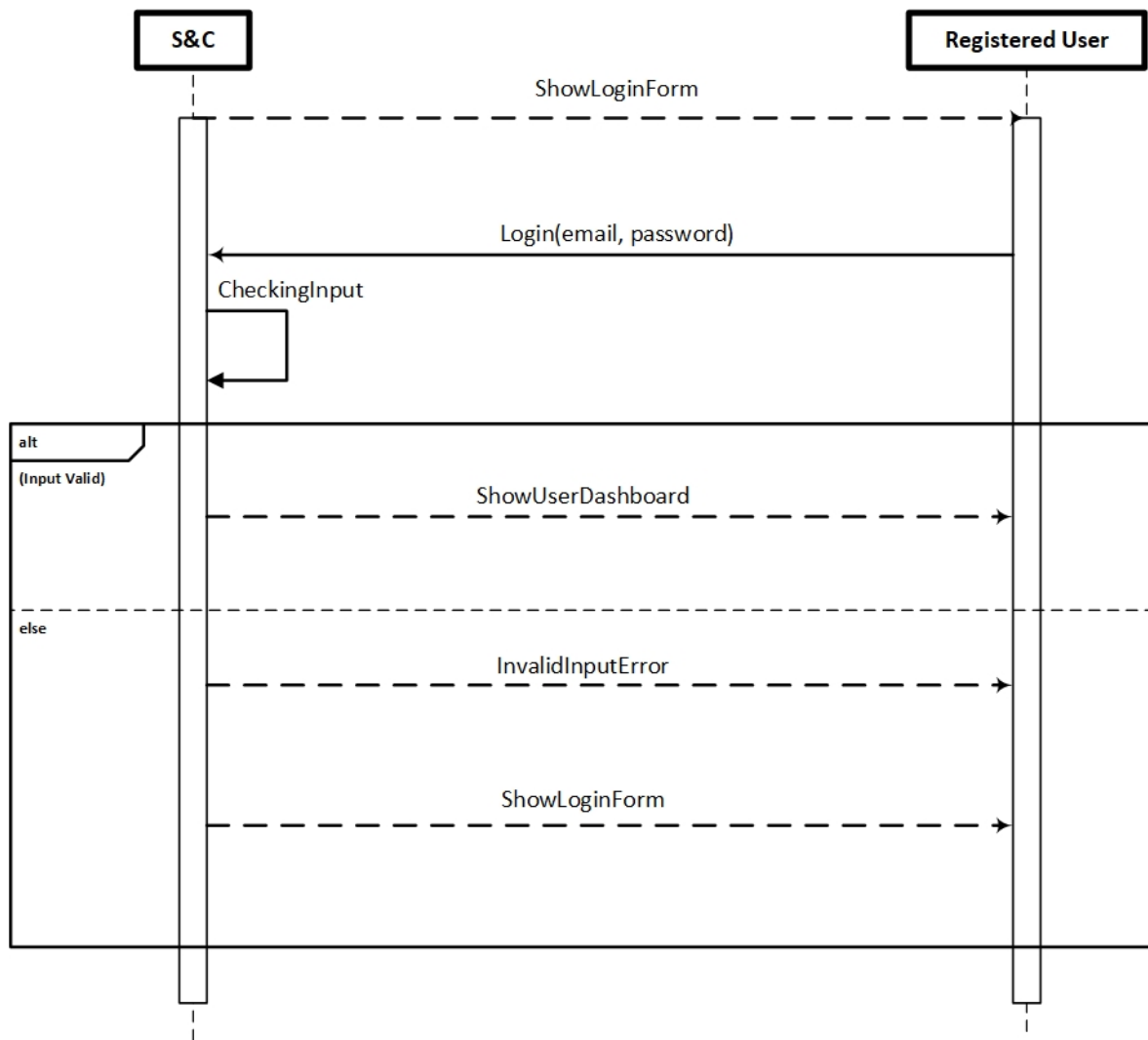


Figure 22: User Login Sequence Diagram

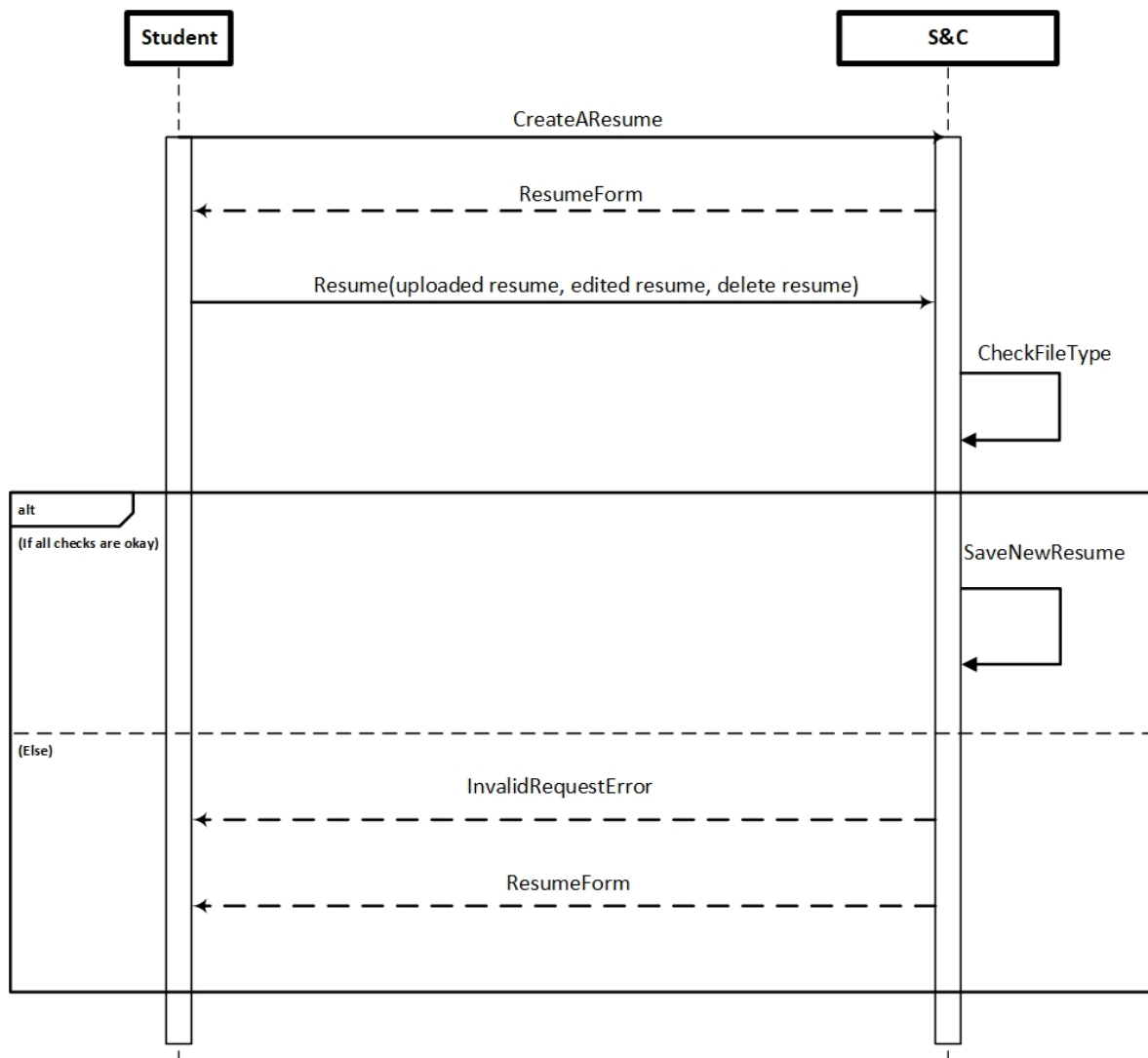


Figure 23: Manage Resume Sequence Diagram

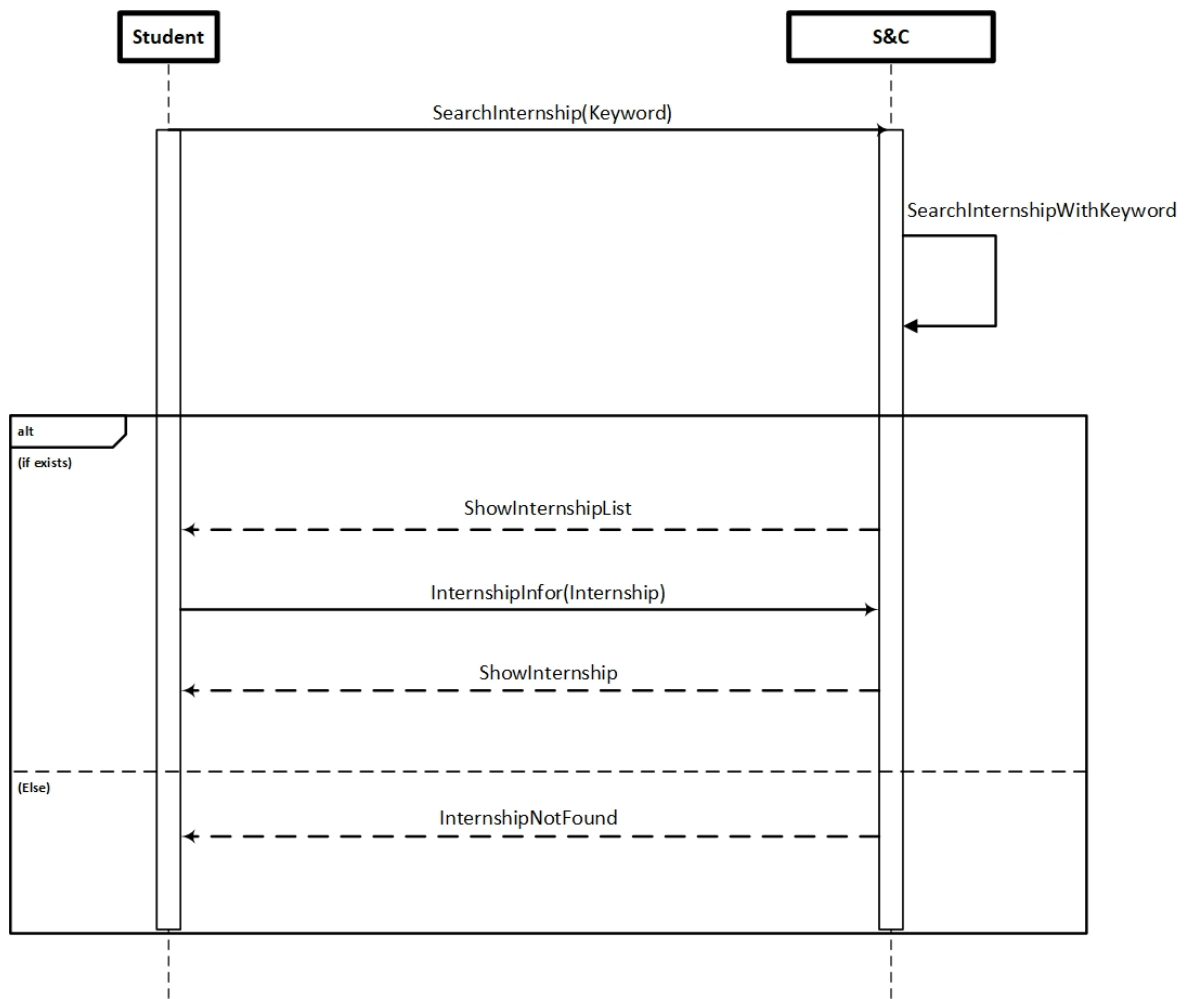


Figure 24: View Internship Sequence Diagram

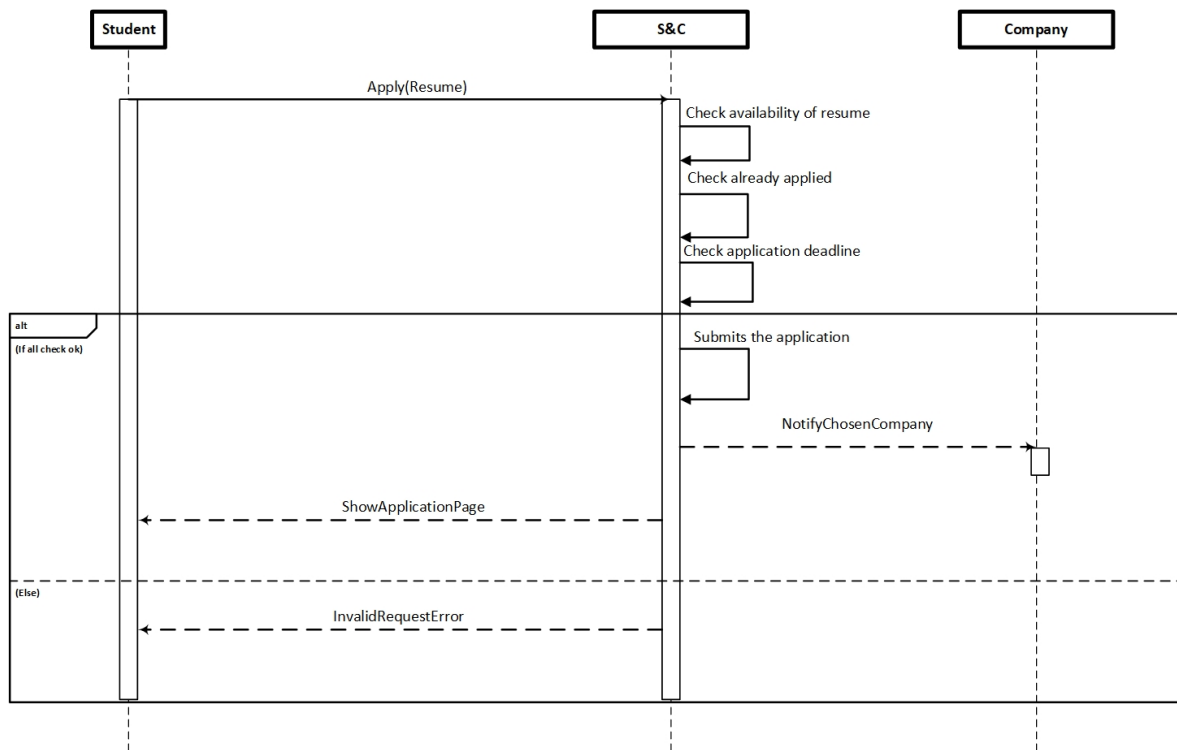


Figure 25: Sending Application Sequence Diagram

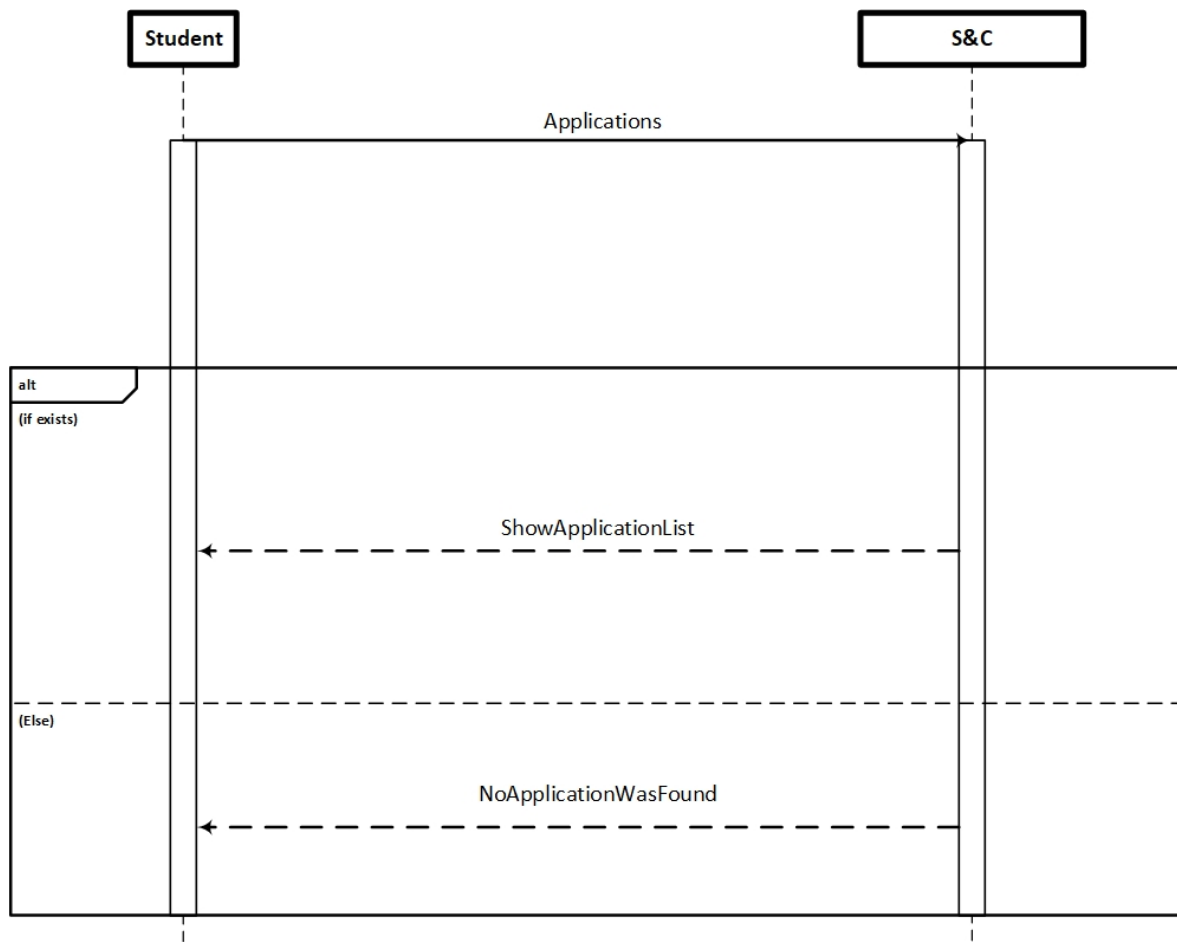


Figure 26: Student Track Applications Sequence Diagram

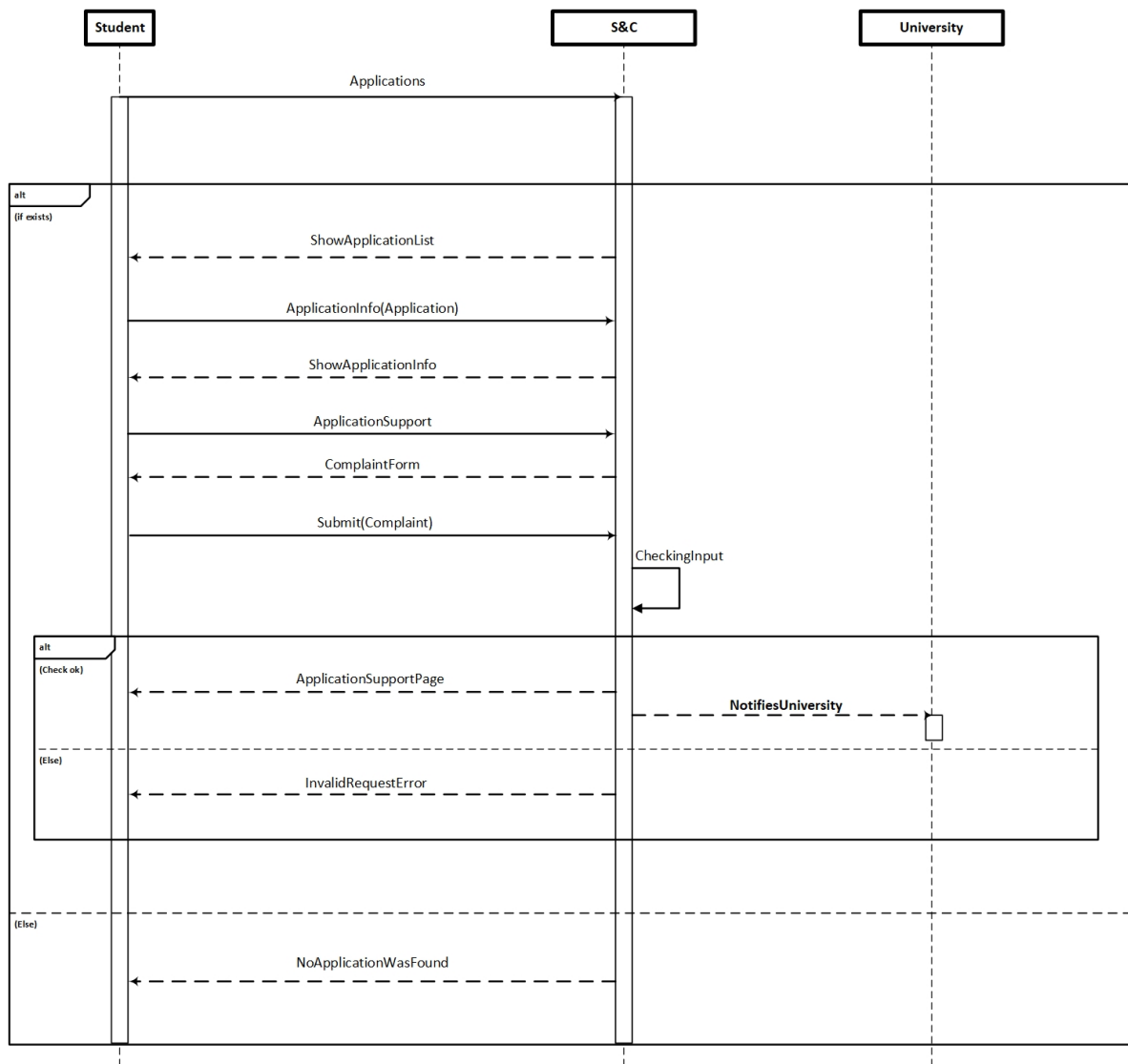


Figure 27: Complaint Submission Sequence Diagram

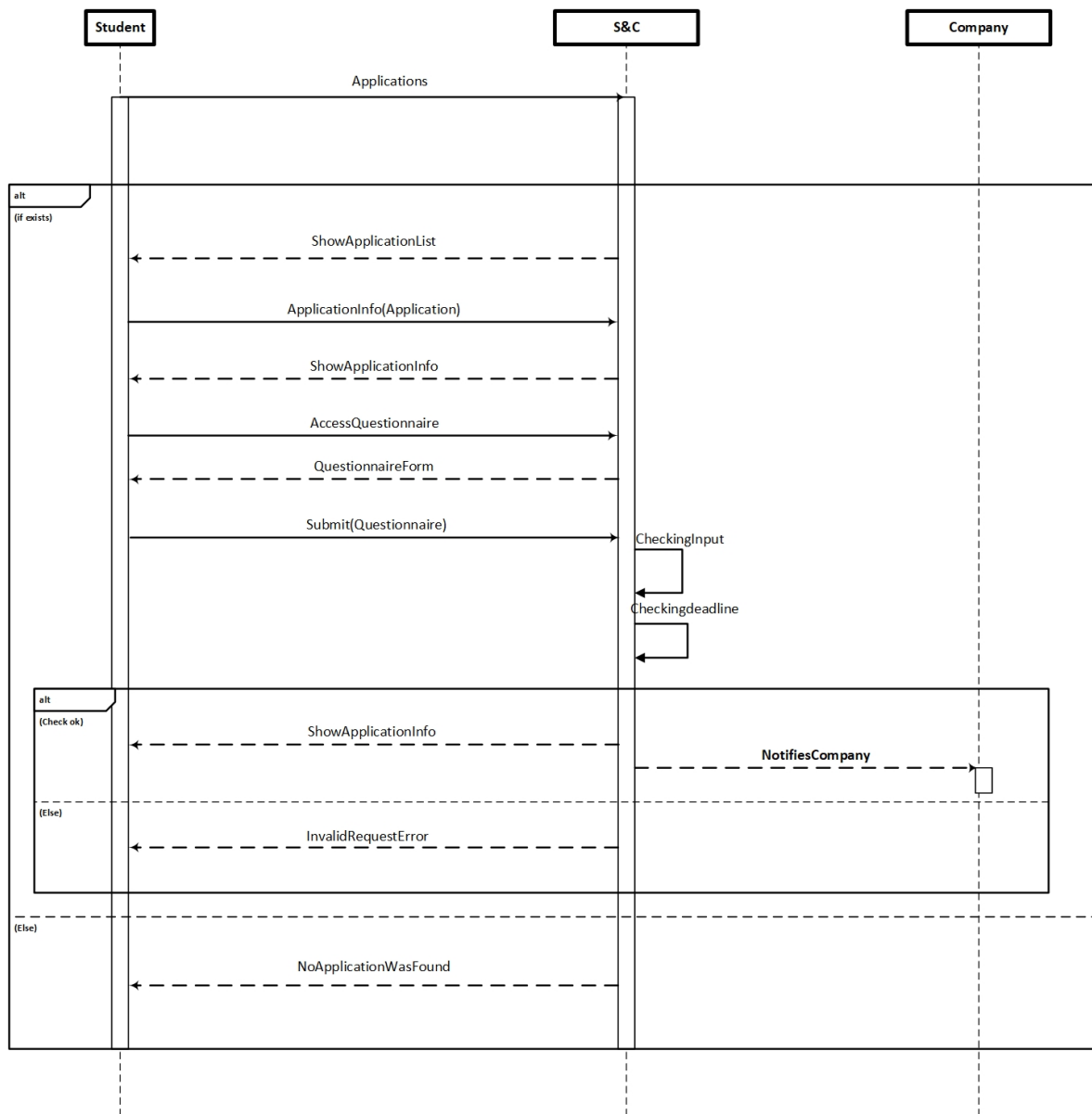


Figure 28: Questionnaire Submission Sequence Diagram

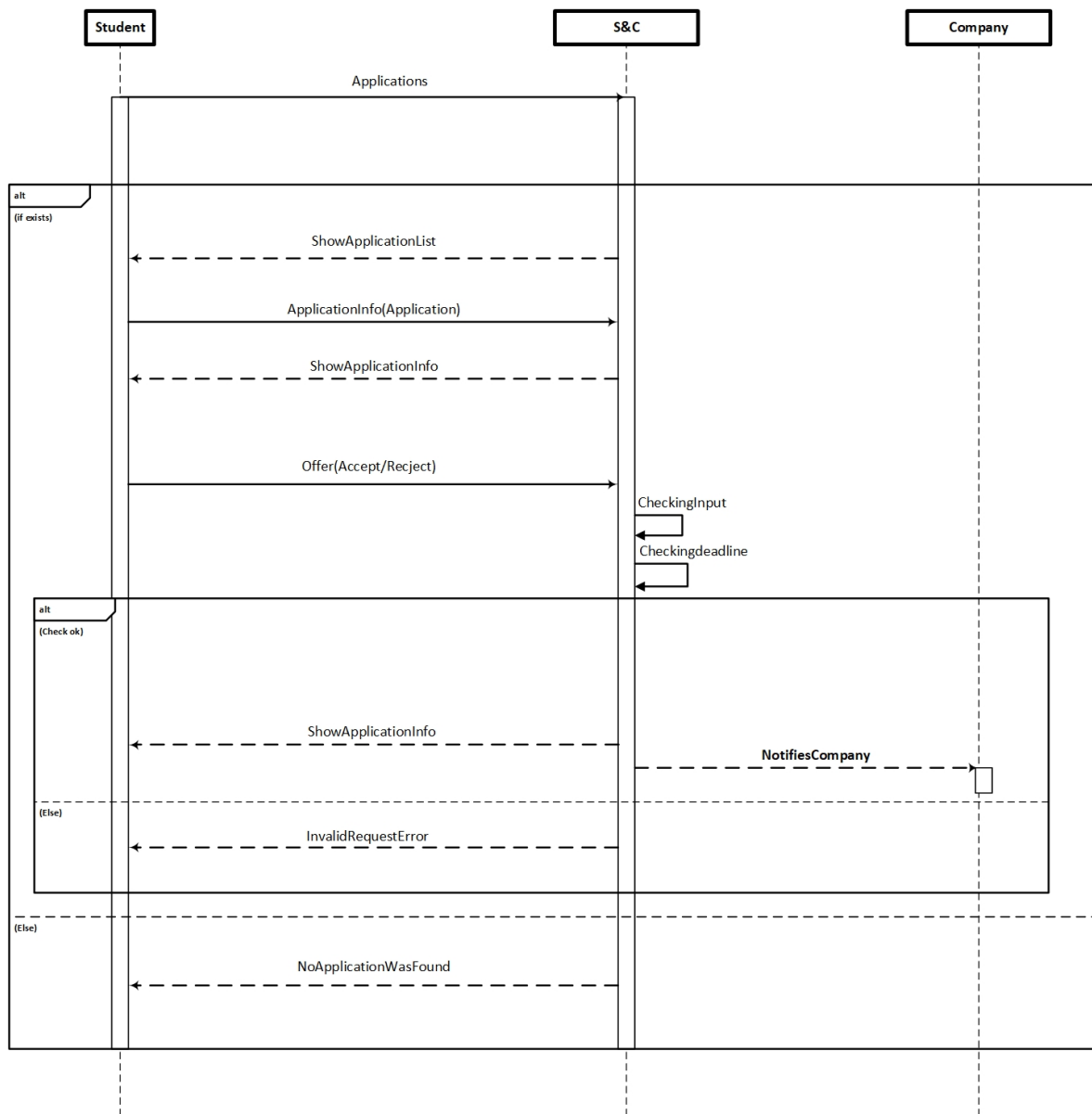


Figure 29: Accept/Reject Offer Sequence Diagram

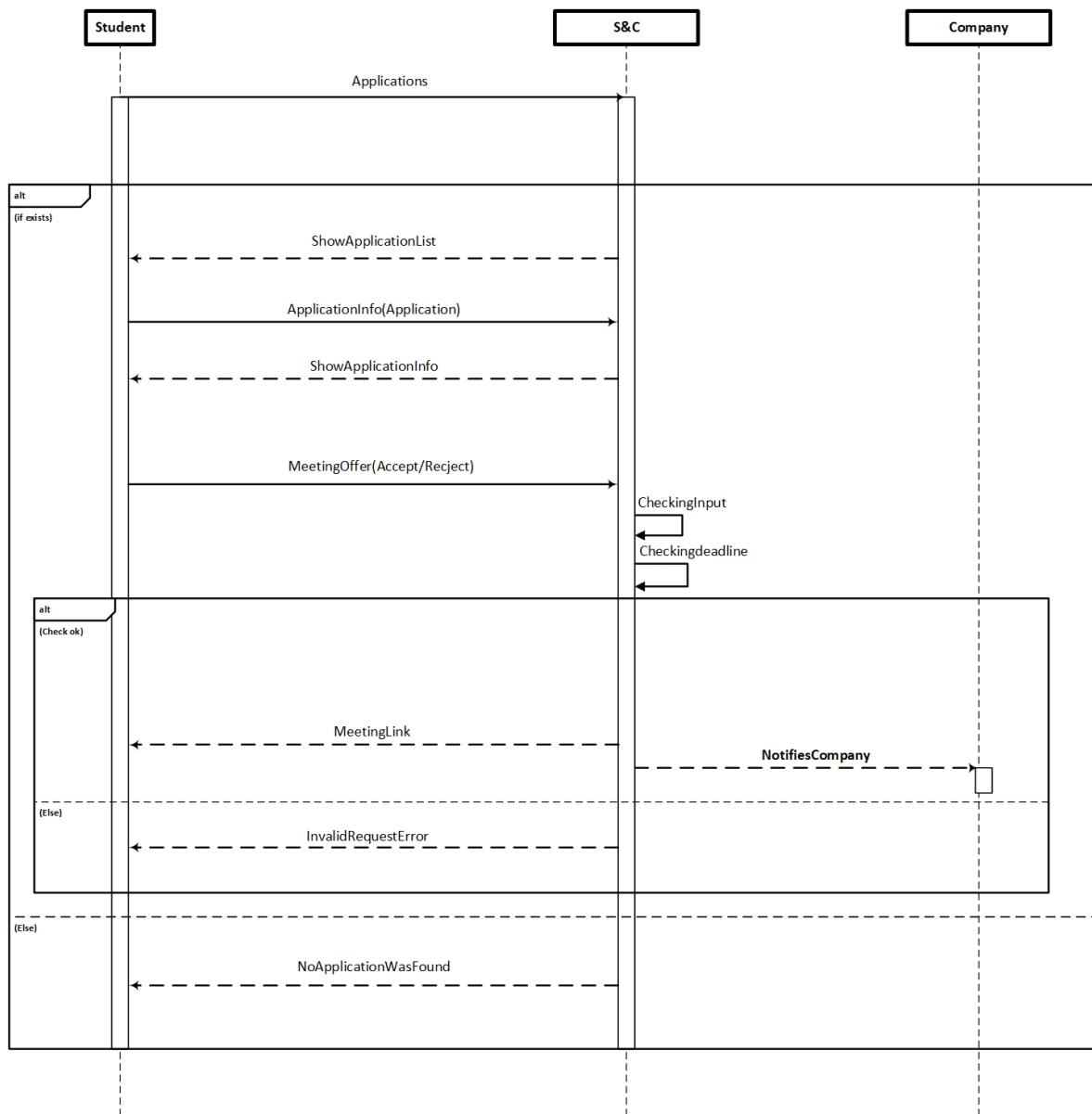


Figure 30: Meeting Attendance Sequence Diagram

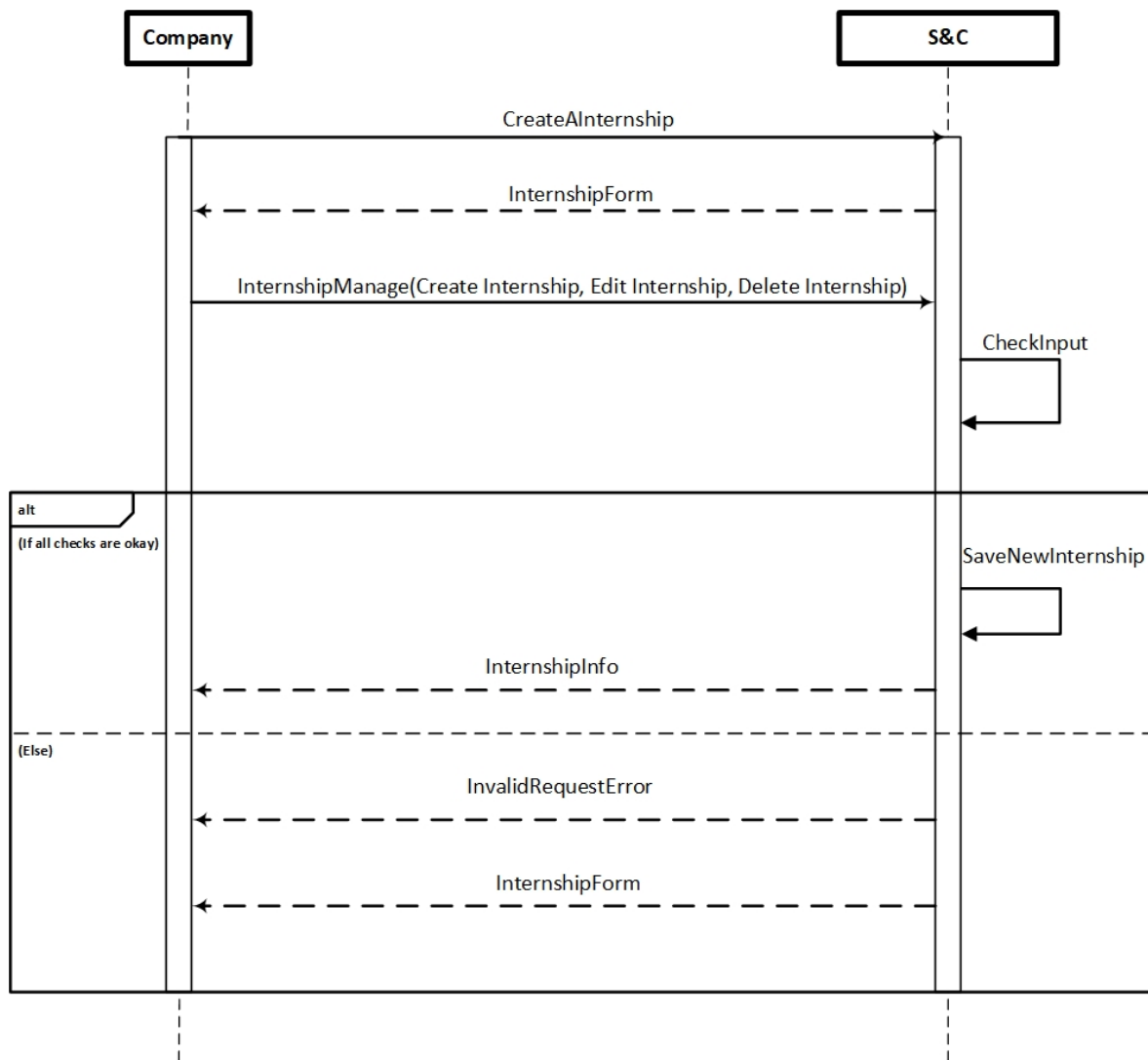


Figure 31: Internship Posting Sequence Diagram

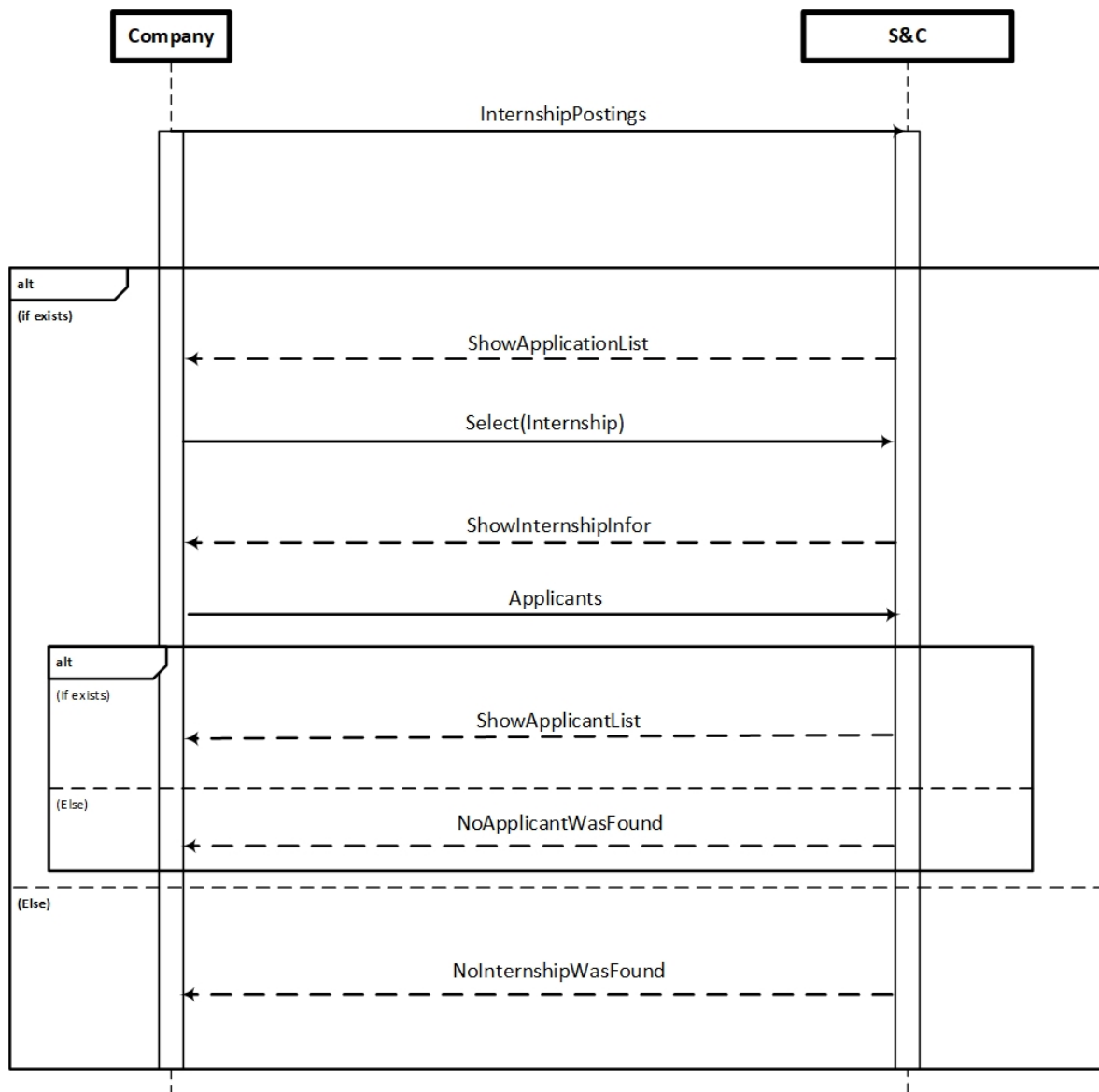


Figure 32: View Applicants Sequence Diagram

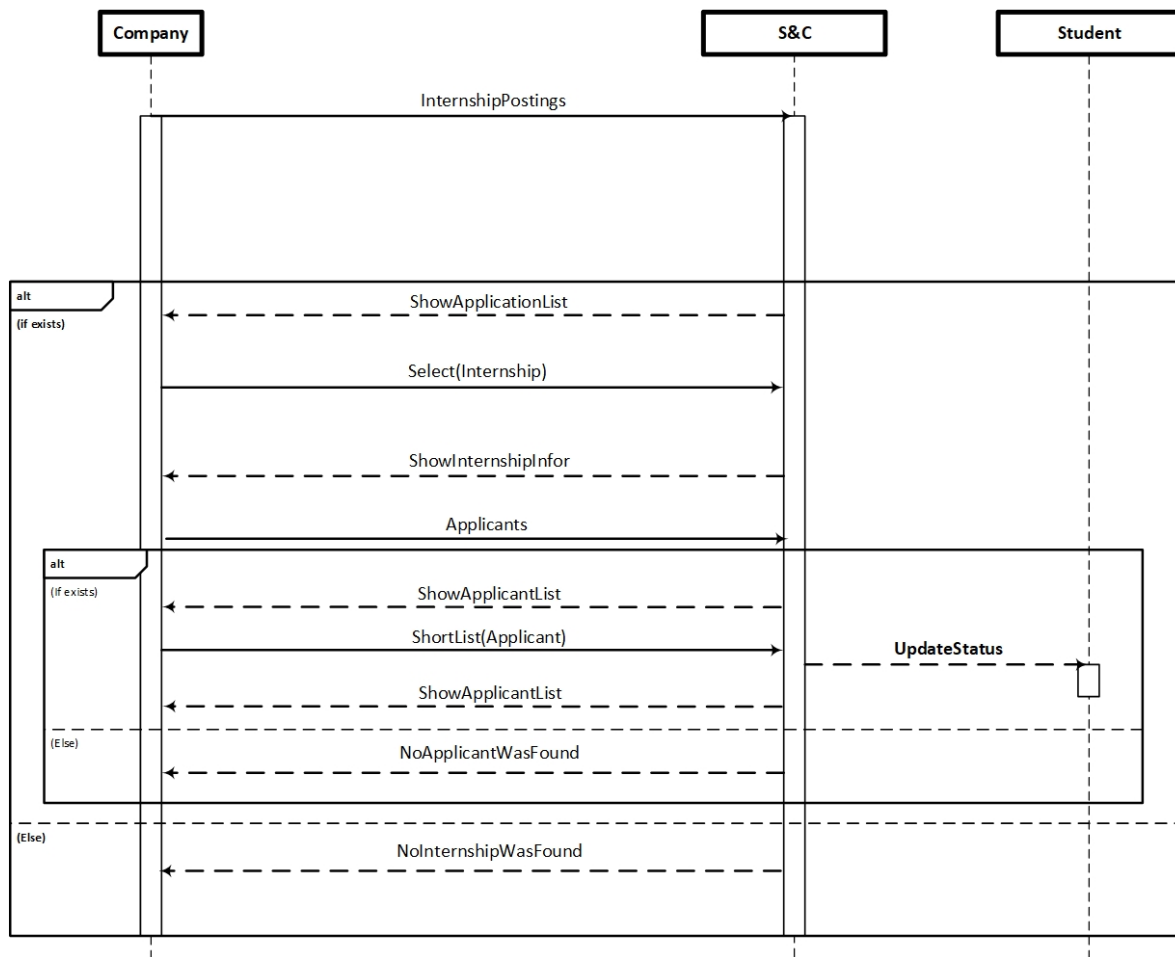


Figure 33: Shortlist Applicants Sequence Diagram

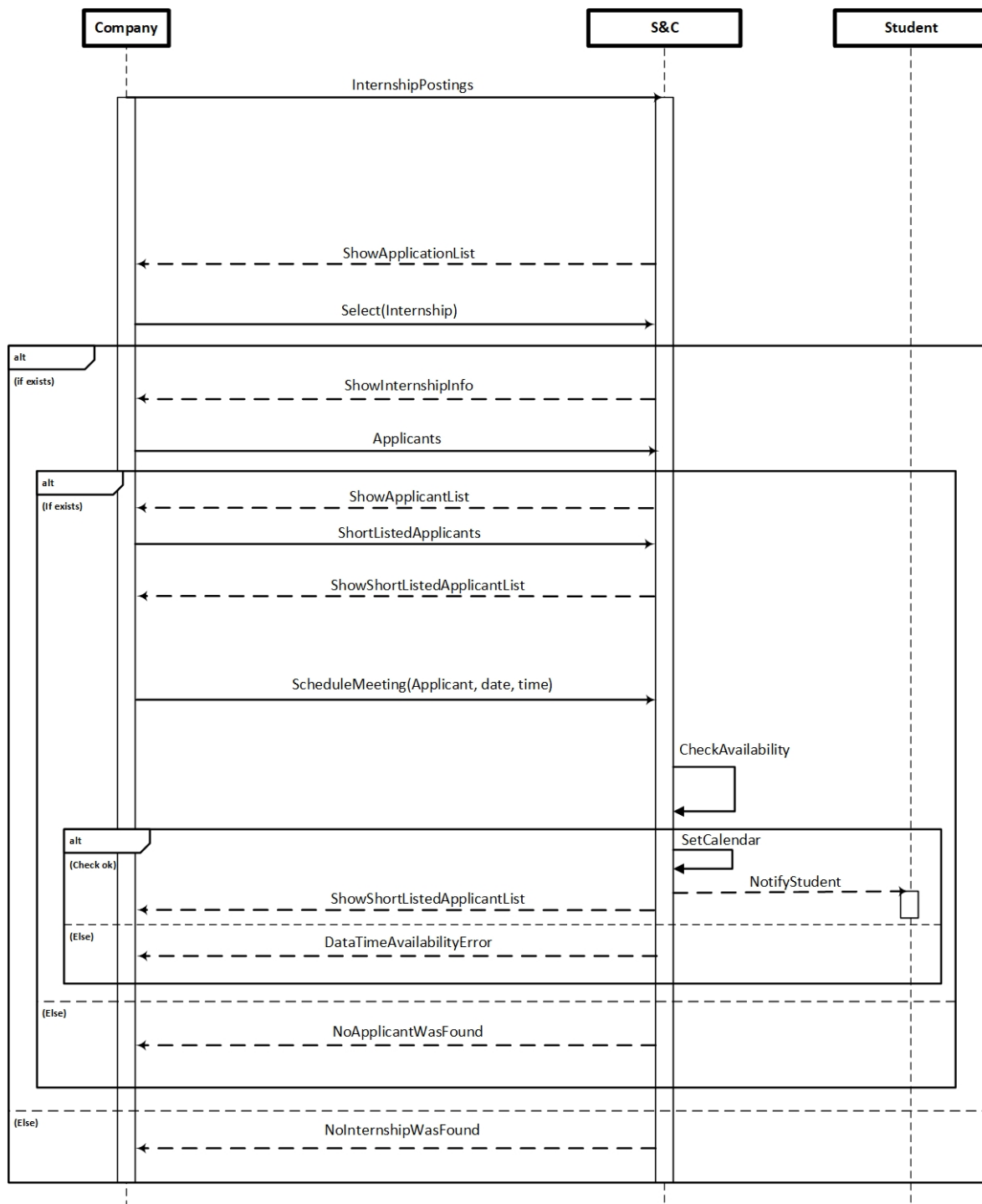


Figure 34: Schedule Meeting Sequence Diagram

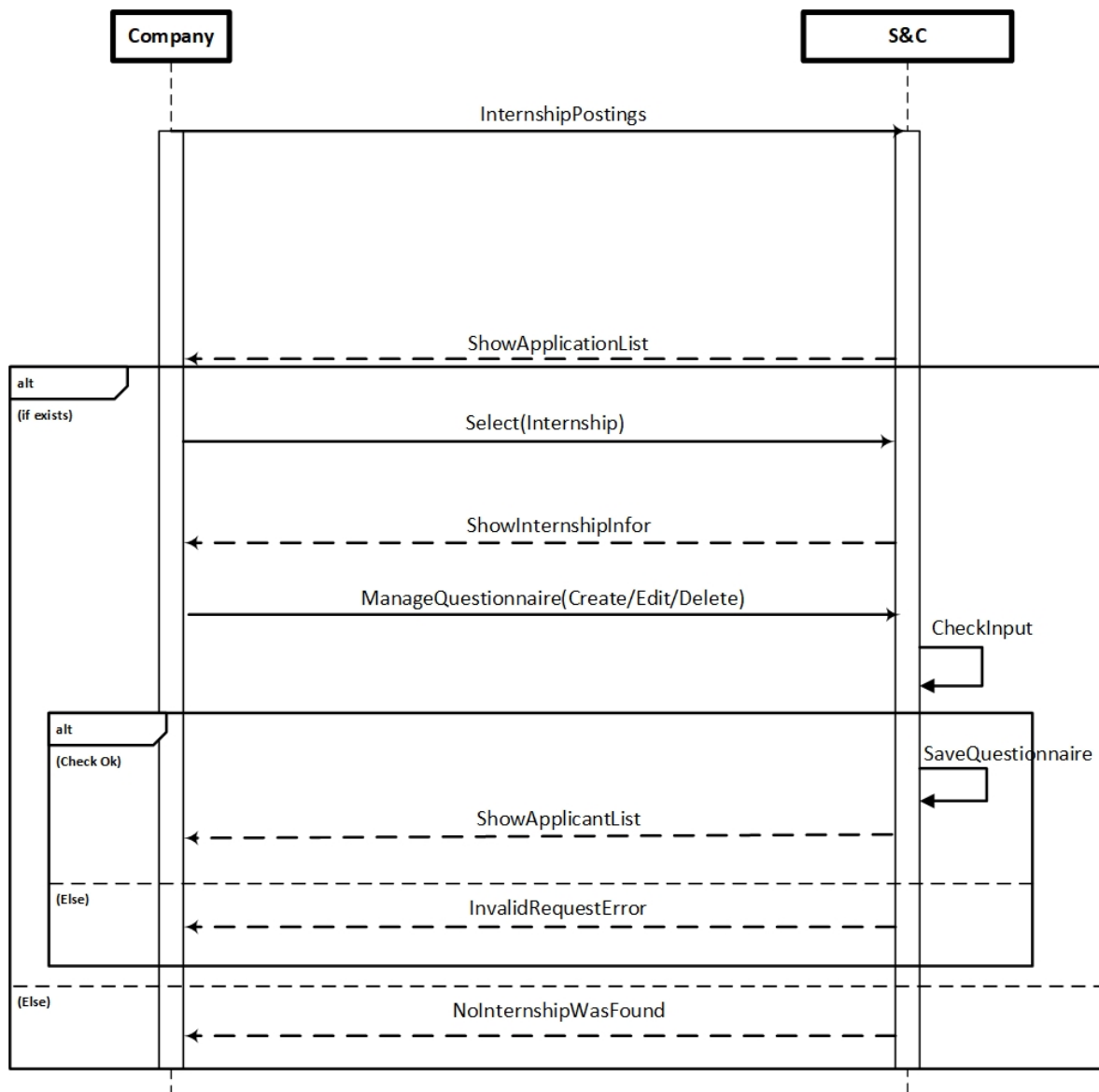


Figure 35: Manage Questionnaire Sequence Diagram

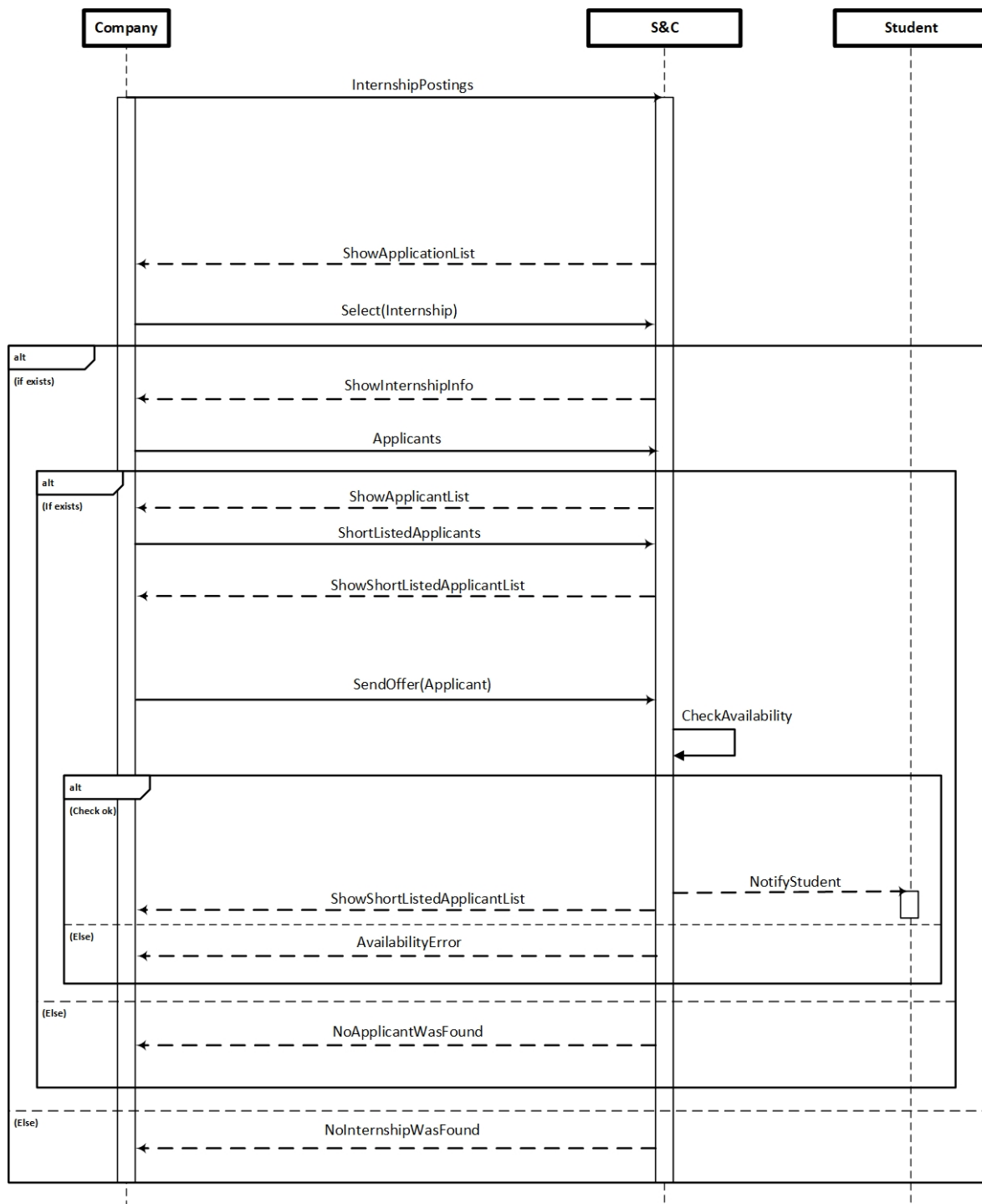


Figure 36: Send Offer Sequence Diagram

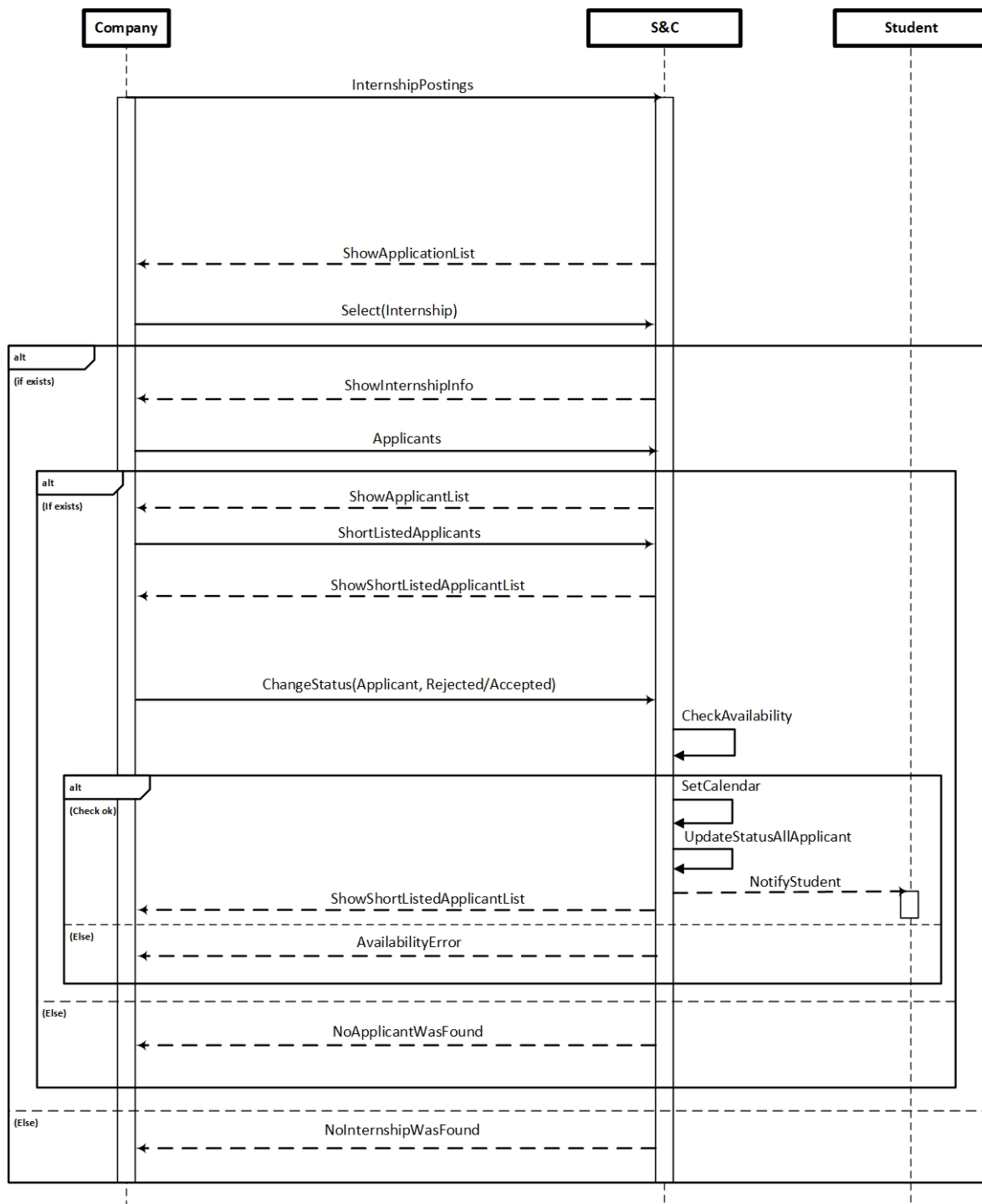


Figure 37: Update Applicant Status Sequence Diagram

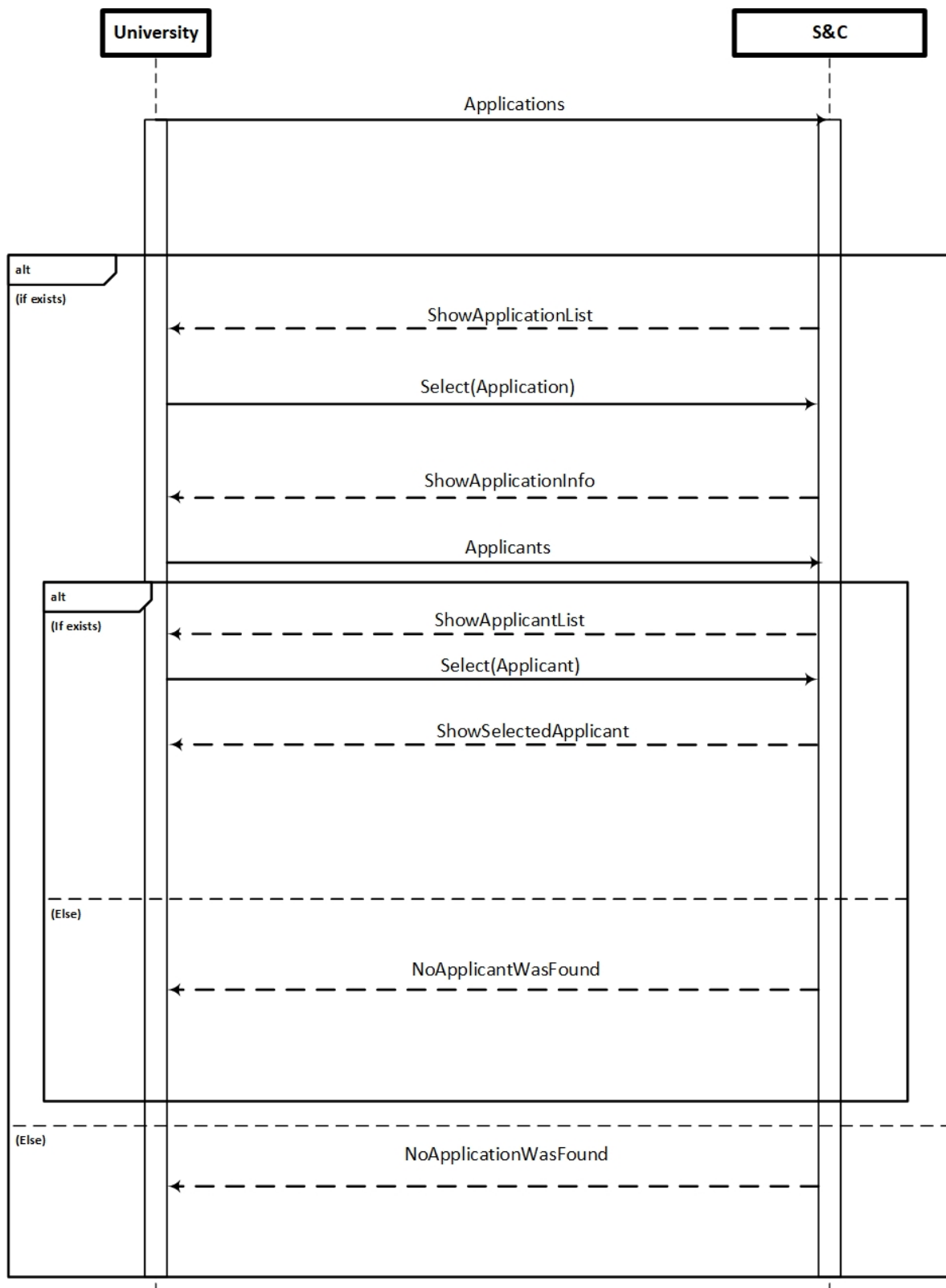


Figure 38: University Track Application Sequence Diagram

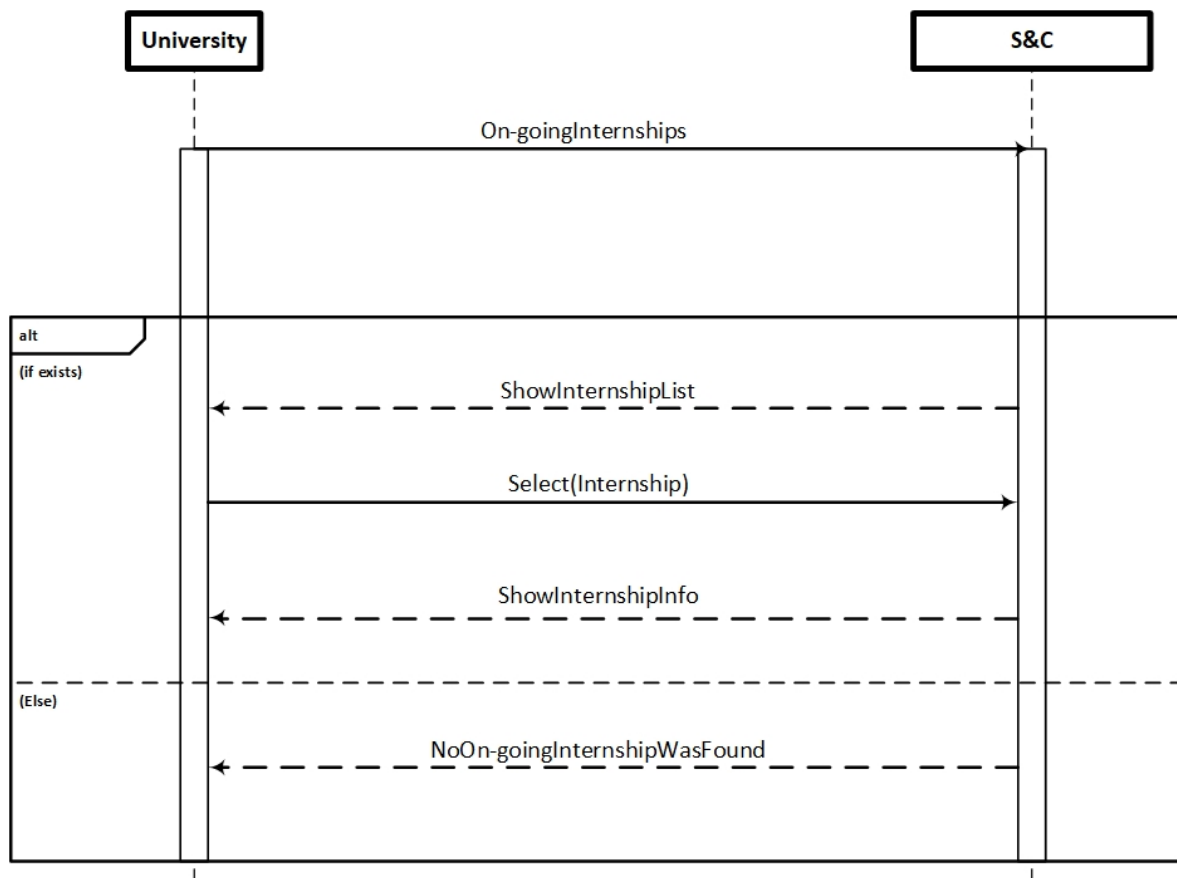


Figure 39: Monitor Internships Sequence Diagram

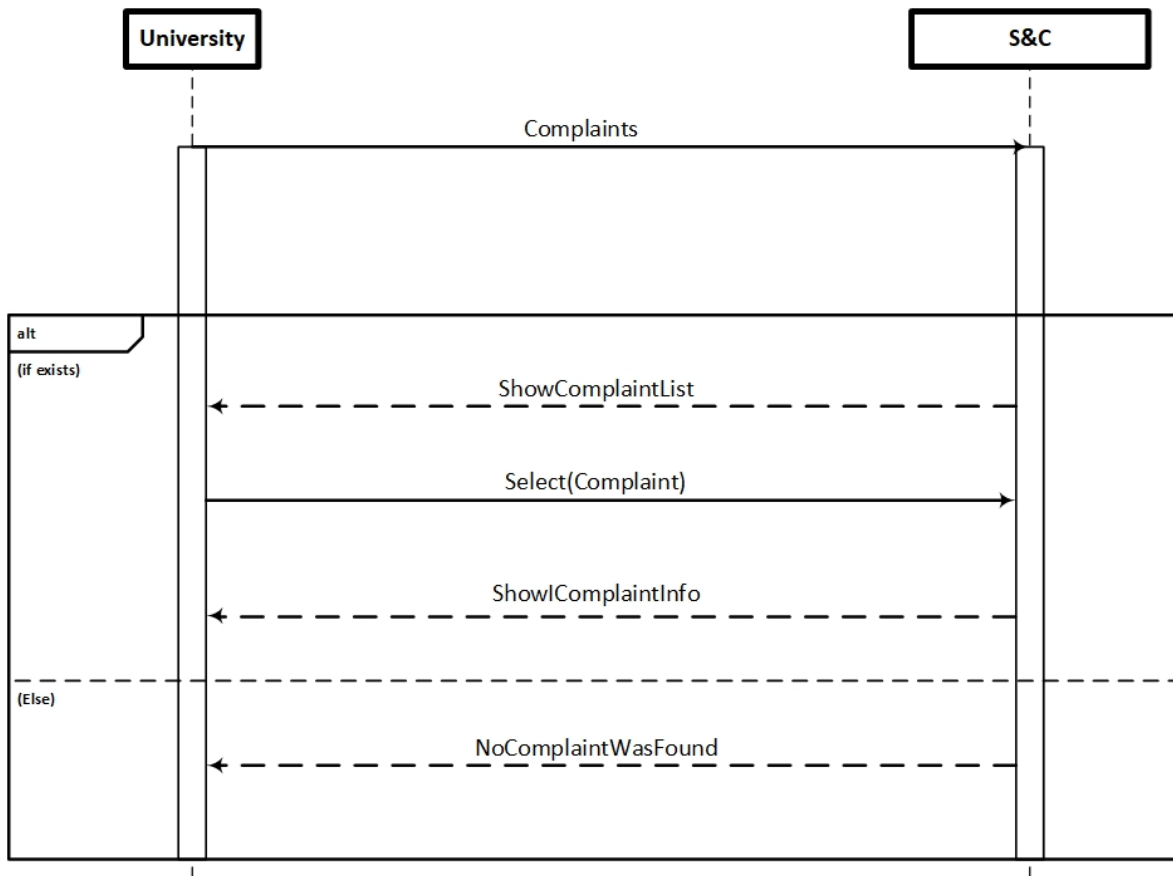


Figure 40: View Complaints Sequence Diagram

3.2.4 Goal Mapping on Requirements

G1 - S&C allows students to manage (create, upload, edit) their resume.

- R1: S&C allows unregistered end users to register.
- R2: After an end user fills all the blanks in registration page correctly, the system must send an email to her/him, in order to confirm her/his email.
- R3: S&C allows logged out Student users to login.
- R6: Users need to have device with stable internet connection.
- R7: S&C allows Student users to upload their resume.
- R8: S&C allows Student users to edit their resume.
- R9: S&C allows Student users to delete their resume.
- R34: S&C allows end users to view their information.
- R39: S&C allows end users to edit their information.
- R40: S&C stores the information about the Users.
- R41: S&C shall ensure security of data.
- R50: S&C must communicate with the mailing system in order to allow Users to register their account

G2 - S&C allows students to see available internships (e.g., by location, industry).

- R6: Users need to have device with stable internet connection.
- R10: S&C allows Student users to search for internship posting.
- R11: S&C allows Student users to view a list internship postings.

G3 - S&C allows students to initiate the search process and submit applications directly

- R10: S&C allows Student users to search for internship posting.
- R11: S&C allows Student users to view a list internship postings.
- R12: S&C allows Student users to send application (resume) for an internship posting.
- R35: S&C must notify Student users to about the submission of application.

G4 - S&C allows students to track the status (reviewed, shortlisted, selected) of their internship applications

- R13: S&C allows Student users to track the status of their applications.
- R14: S&C allows Student users to view a list of their applications.
- R36: S&C must notify Student users to about the change of application's status.

G5 - S&C allows students to raise complaints or concerns regarding an internship experience

- R10: S&C allows Student users to search for internship posting.
- R11: S&C allows Student users to view a list internship postings.
- R12: S&C allows Student users to send application (resume) for an internship posting.
- R14: S&C allows Student users to view a list of their applications.
- R15: S&C allows Student users to submit complaints about internship they are involved in to their university.
- R40: S&C stores the information about the Users.
- R41: S&C shall ensure security of data.
- R44: S&C must notify University users to about the receiving complaints from students involved in internship

G6 - S&C allows companies to manage (create, edit, publish, delete) internship listings

- R1: S&C allows unregistered end users to register.
- R2: After an end user fills all the blanks in registration page correctly, the system must send an email to her/him, in order to confirm her/his email.
- R4: S&C allows logged out Company users to login.
- R20: S&C allows Company users to enter information for internship offerings
- R21: S&C allows Company users to post/publish internship offerings.
- R22: S&C allows Company users to edit posted internship offerings.
- R23: S&C allows Company users to delete posted internship offerings.
- R39: S&C allows end users to edit their information.
- R40: S&C stores the information about the Users.
- R41: S&C shall ensure security of data.
- R50: S&C must communicate with the mailing system in order to allow Users to register their account.

G7 - S&C allows companies to browse and search through student profiles (CVs) applied for the internship (e.g., by location, industry)

- R24: S&C allows Company users to view all the applicant's application for an posted internship offering.

G8 - S&C allows companies the ability to shortlist candidates

- R24: S&C allows Company users to view all the applicant's application for an posted internship offering.
- R25: S&C allows Company users to shortlist applicants for an posted internship offering.

G9 - S&C allows companies communicate with applicants

- R31: S&C allows Company users to send proposal/offer for an posted internship offering.
- R42: S&C must notify Company users to about the acceptance/rejection of request for meeting from an applicant.
- R43: S&C must notify Company users to about the acceptance/rejection of the offering internship proposal from an applicant.
- R16: S&C allows Student users to submit questionnaire assessment test for an internship.
- R18: S&C allows Student users to accept/reject internship offerings.

G10 - S&C allows companies to provide updates on their application status of incoming applications

- R24: S&C allows Company users to view all the applicant's application for an posted internship offering.
- R25: S&C allows Company users to shortlist applicants for an posted internship offering.
- R31: S&C allows Company users to send proposal/offer for an posted internship offering.
- R46: S&C must update the status of Student users application once a new change has been placed.
- R48: S&C must update the calendar of Student users for internship periods.
- R36: S&C must notify Student users to about the change of application's status.

G11 - S&C allows companies to manage (create, edit, send) questionnaire to applicants

- R27: S&C allows Company users to create questionnaire for an posted internship offering.
- R28: S&C allows Company users to send questionnaire for an posted internship offering.
- R29: S&C allows Company users to edit questionnaire for an posted internship offering.
- R30: S&C allows Company users to delete questionnaire for an posted internship offering.
- R38: S&C must notify Company users to about the submission of questionnaire to an applied internship
- R39: S&C allows end users to edit their information.
- R40: S&C stores the information about the Users.
- R41: S&C shall ensure security of data.
- R16: S&C allows Student users to submit questionnaire assessment test for an internship.

G12 - S&C allows companies to manage (scheduling and conducting) interviews with applicants

- R25: S&C allows Company users to shortlist applicants for an posted internship offering.
- R26: S&C allows Company users to send meeting proposal to shortlisted applicant for an posted internship offering.
- R42: S&C must notify Company users to about the acceptance/rejection of request for meeting from an applicant.
- R47: S&C must update the calendar of Student users for meetings.
- R49: S&C must update the calendar of Company users for meetings.
- R37: S&C must notify Student users to about the request for meeting from an applied internship.
- R19: S&C allows Student users to accept/reject meeting offerings.
- R17: S&C allows Student users to take part in the meetings.

- R39: S&C allows end users to edit their information.
- R40: S&C stores the information about the Users.
- R41: S&C shall ensure security of data.

G13 - S&C allows universities to track application processes

- R1: S&C allows unregistered end users to register.
- R2: After an end user fills all the blanks in registration page correctly, the system must send an email to her/him, in order to confirm her/his email.
- R5: S&C allows logged out University users to login
- R51: S&C allows University users to search for and view a list of their students.
- R32: S&C allows University users to track applications for an posted internship offering involved their students.
- R45: S&C must notify University users to about the change of status of applications from students involved in internship.
- R39: S&C allows end users to edit their information.
- R40: S&C stores the information about the Users.
- R41: S&C shall ensure security of data.
- R50: S&C must communicate with the mailing system in order to allow Users to register their account.

G14 - S&C allows universities to monitor internship execution

- R1: S&C allows unregistered end users to register.
- R2: After an end user fills all the blanks in registration page correctly, the system must send an email to her/him, in order to confirm her/his email.
- R5: S&C allows logged out University users to login
- R51: S&C allows University users to search for and view a list of their students.
- R53: S&C allows University users to search for and monitor internship involved with their students
- R54: S&C must notify University users to about the change of status of applications from students involved in internship.
- R39: S&C allows end users to edit their information.
- R40: S&C stores the information about the Users.
- R41: S&C shall ensure security of data.
- R50: S&C must communicate with the mailing system in order to allow Users to register their account.

G15 - S&C allows universities to collect complaints

- R1: S&C allows unregistered end users to register.
- R2: After an end user fills all the blanks in registration page correctly, the system must send an email to her/him, in order to confirm her/his email.
- R5: S&C allows logged out University users to login
- R51: S&C allows University users to search for and view a list of their students.
- R52: S&C allows University users to search for and view a list of complaints from their students.
- R15: S&C allows Student users to submit complaints about internship they are involved in to their university.
- R44: S&C must notify University users to about the receiving complaints from students involved in internship.
- R40: S&C stores the information about the Users.
- R41: S&C shall ensure security of data.
- R33: S&C allows University users to receive complaints for an posted internship offering involved their students.
- R50: S&C must communicate with the mailing system in order to allow Users to register their account.

Requirements	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15
R1	✓	✓				✓							✓	✓	✓
R2	✓					✓							✓	✓	✓
R3	✓														
R4						✓									
R5													✓	✓	✓
R6	✓	✓													
R7	✓														
R8	✓														
R9	✓														
R10		✓	✓		✓										
R11		✓	✓		✓										
R12			✓		✓										
R13				✓											
R14				✓	✓										
R15					✓										✓
R16									✓						
R17												✓			
R18									✓						
R19									✓			✓			
R20						✓									
R21						✓									
R22						✓									
R23						✓									
R24							✓	✓		✓					
R25								✓		✓		✓			
R26												✓			
R27											✓				
R28											✓				
R29											✓				
R30											✓				
R31									✓	✓					
R32													✓		
R33															✓
R34	✓														
R35			✓						✓						
R36				✓						✓					

R37												✓	✓			
R38												✓				
R39	✓					✓						✓	✓	✓	✓	
R40	✓					✓	✓					✓	✓	✓	✓	✓
R41	✓					✓	✓					✓	✓	✓	✓	✓
R42									✓				✓			
R43									✓							
R44						✓										✓
R45														✓		
R46										✓						
R47													✓			
R48										✓						
R49													✓			
R50	✓						✓							✓	✓	✓
R51														✓	✓	✓
R52																✓
R53															✓	
R54															✓	

Table 31: Traceability Matrix for Goals and Requirements

3.3 Performance Requirements

- The system must be able to serve a great number of users simultaneously.
- The system must guarantee correct responses.
- The system must be able to send a response to a query less than 3 seconds since it has been received.
- The system must be available 99% of the time.

3.4 Design Constraints

3.4.1 Standard Compliance

The system must comply with the EU's GDPR (General Data Protection Regulation), which protects personal data, privacy, and security for EU citizens.

3.4.2 Hardware Limitations

The only hardware constraints are the need for a stable internet connection and a web browser.

3.4.3 Any Other Constraint

The system has to be fault tolerant in order to prevent the propagation of errors and to guarantee a continuous usability of the system.

3.5 Software System Attributes

3.5.1 Reliability

The system must be able to run continuously without any interrupts. Reliability of the system depends on the services of the system, and should be up for a 99% of time. This means the MTTR or downtime should be 3.65 days per years. In order to guarantee this time of downtime the system must have an appropriate infrastructure with a fully backup system located in different office that replicates the core services for covering general failure of the main system.

3.5.2 Availability

The system must be available the most time possible, with a minimum value of 99.9%(three-nines) of time. In this way the system will be unavailable for only 8.76 hours a year. It shall be prevented a case scenario in which a maintenance break occurs near to application deadline.

3.5.3 Security

The system must control the access rights of the Users. The system shall grant both authentication, verifying the identity of the Users that attempt to login and authorization, verifying the permission of the already logged Users to perform certain requested actions. Measures to protect the database will be adopted, such as defense against query injections, and password and Users' personal data stored will be encrypted.

Since the S&C system keeps the personal and resume data, the security is an important aspects of the system.

3.5.4 Maintainability

The system must be designed using scalable and reusable models in order to permit future addition of features with minimum effort. Ordinary maintenance has to be scheduled at nigh-time, in order to keep the services available when the User traffic is high.

3.5.5 Portability

The system must be accessible by the Users from every kind of Web Browser. There are no particular portability requirements server side.

4 Formal Analysis using Alloy

```
sig Firstname{}
sig Lastname{}
sig Email{}
sig Password{}
sig CompanyName{}
sig UniversityName{}
sig Resume{}
sig Time {
    value: one Int
}

abstract sig Status{}
one sig ACCEPT extends Status{}
one sig REJECT extends Status{}
one sig SHORTLIST extends Status{}
one sig SUBMITTED extends Status{}

abstract sig User {
    name: one Firstname,
    lastname: one Lastname,
    email: one Email
}

sig Student extends User {
    university: one UniversityName,
    password: one Password,
    resume: one Resume,
    meetings: set Meeting,
    applications: set Application,
```

```

    complaints: set Complaint
}

sig Company extends User {
    password: one Password,
    meetings: set Meeting,
    internships: set Internship,
    questionnaires: set Questionnaire
}

sig University extends User {
    nameUniversity: one UniversityName,
    password: one Password,
    students: set Student,
    monitoredInternships: set Internship,
    monitoredApplications: set Application,
    receivedComplaints: set Complaint
}

sig Internship {
    company: one Company,
    applicants: set Student,
    deadline: one Time,
    acceptedStudent: lone Student
}

sig Application {
    student: one Student,
    internship: one Internship,
    status: one Status,
    timeSubmitted: one Time
}

sig Complaint {
    student: one Student,
    university: one University,
    application: one Application,
    timeSubmitted: one Time
}

sig Questionnaire {
    internship: one Internship,
    assignedStudents: set Student,
    submissions: set Student
}

sig Meeting {
    student: one Student,
    internship: one Internship,
    deadline: one Time
}

fact ApplicationDeadline {
    all a: Application | a.internship.deadline in Time
}

```

```

fact UniqueActiveInternship {
  all s: Student | lone i: Internship | i.acceptedStudent = s
}

fact ValidComplaints {
  all c: Complaint | c.application in c.student.applications
}

fact QuestionnaireConstraints {
  all q: Questionnaire | q.assignedStudents in q.internship.applicants
}

fact ValidApplication {
  all a: Application |
    a.internship.deadline.value >= a.timeSubmitted.value

  all a: Application |
    one a.student and one a.internship
}

fact UniqueInternshipPerCompany {
  all i: Internship | lone c: Company | i in c.internships
}

fact UniqueMeetingTimesPerInternship {
  all i: Internship, m1, m2: Meeting |
    m1 != m2 and m1.internship = i and m2.internship = i =>
      m1.deadline != m2.deadline
}

fact UniqueMeetingTimesPerStudent {
  all s: Student, m1, m2: Meeting |
    m1 != m2 and m1.student = s and m2.student = s =>
      m1.deadline != m2.deadline
}

fact ApplicationsBelongToInternship {
  all a: Application | a in a.internship.applicants.applications
}

fact ValidComplaintSubmission {
  all c: Complaint |
    c.application.student = c.student and
    c.university.nameUniversity = c.student.university
}

fact ApplicationAssociatedToOneStudent {
  all a: Application | one a.student
}

fact UniversityMonitorsOnlyOwnApplications {
  all u: University, a: Application |
    a in u.monitoredApplications => a.student in u.students
}

fact UniversityMonitorsOnlyOwnInternships {

```

```

    all u: University, i: Internship |
      i in u.monitoredInternships => i.applicants in u.students
  }

fact ValidQuestionnaireSubmission {
  all q: Questionnaire, s: Student |
    s in q.assignedStudents => s in q.internship.applicants
}

fact UniqueEmailAcrossUsers {
  all u1, u2: User |
    u1 != u2 => u1.email != u2.email
}

fact UniqueApplicationPerInternship {
  all i1, i2: Internship, a: Application |
    i1 != i2 => (a in i1.applicants.applications and a in i2.applicants.applications)
      implies i1 = i2
}

fact NoStudentAcceptedByTwoInternships {
  all s: Student, i1, i2: Internship |
    i1 != i2 and i1.acceptedStudent = s and i2.acceptedStudent = s => i1 = i2
}

fact NoStudentFromTwoUniversities {
  all s: Student, u1, u2: University |
    u1 != u2 and s.university = u1.nameUniversity and s.university = u2.nameUniversity
      => u1 = u2
}

fact NoDuplicateResumes {
  all s1, s2: Student |
    s1 != s2 => s1.resume != s2.resume
}

fact applicationAcceptedStatus {
  all s: Student, i: Internship, a: Application |
    i.acceptedStudent = s and
    a.student = s and
    a.internship = i
    => a.status = ACCEPT
}

fact internshipAcceptanceAndRejection {
  all i: Internship, s: Student, a: Application |
    i.acceptedStudent = s and
    a.student = s and a.internship = i
    => a.status = ACCEPT

  all i: Internship, s: Student, a: Application |
    i.acceptedStudent = s and
    a.student != s and a.internship = i
    => a.status = REJECT
}

```

```

}

pred PostInternship(c: Company, i: Internship) {
  i.company = c
  i.deadline.value > 0
  no i.applicants
  no i.acceptedStudent
}

pred AcceptInternshipApplication(s: Student, i: Internship, a: Application) {
  a.status = ACCEPT
  i.acceptedStudent = s

  all a1: Application |
    a1.internship = i and a1 != a and a1.student != s =>
      a1.status = REJECT

  no i.applicants - s
}

pred CheckDuplicateResumes(s1: Student, s2: Student) {
  s1 != s2 => s1.resume != s2.resume
}

pred ValidateEmailUniqueness(u1: User, u2: User) {
  u1 != u2 => u1.email != u2.email
}

pred StudentAcceptedByOneInternship(s: Student) {
  lone i: Internship | i.acceptedStudent = s
}

pred CheckApplicationDeadline(a: Application) {
  a.internship.deadline.value >= a.timeSubmitted.value
}

pred MonitorInternships(u: University, i: Internship) {
  i.applicants in u.students
  i in u.monitoredInternships
}

pred MonitorApplications(u: University, a: Application) {
  a.student in u.students
  a in u.monitoredApplications
}

pred SubmitQuestionnaire(q: Questionnaire, s: Student) {
  s in q.assignedStudents
  s in q.submissions
}

pred AssignQuestionnaire(q: Questionnaire, i: Internship, students: set Student) {
  q.internship = i
  q.assignedStudents = students
}

```

```
    students in i.applicants
}

pred world1 {
    #Student > 5
    #University> 3
    #Company > 1
    #Internship > 2
    #Application > 3
    #Meeting > 1
}

run world1 for 10
```

Executing "Run world1 for 10"

Solver=sat4j Bitwidth=4 MaxSeq=7 SkolemDepth=1 Symmetry=20 Mode=batch
74242 vars. 3960 primary vars. 142924 clauses. 250ms.

Instance found. Predicate is consistent. 223ms.

Figure 41: Alloy Execution Example.



5 Effort Spent

Table 32: Effort spent by member of the group

Member of Group	Effort spent
Ali Noshad	• Introduction: 5h • Overall description: 22h • Specific Requirements: 37h • Formal Analysis: 14h • Reasoning: 8h

References

- ISO/IEC/IEEE 29148:2018 - Systems and software engineering - Life cycle processes - Requirements engineering.
- Course Slides
- The Requirement Engineering and Design Project specification document A.Y. 2024-2025.

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